EC24101 BASIC ELECTRONICS

Module 1: Diodes and Applications

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Overview

Introduction to Electronics

- The word "Electronics" is derived from electron mechanics which means the study of the behavior of an electron under different conditins of externally applied fields.
- It is that field of science and engineering, which deals with electron devices and their utilization.
- An electron devices is a device in which conduction takes place by the movement of electrons-through a vaccum, a gas, or a semiconductor.

Applications of Electronics

Electronics play a major role in almost every sphere of our life:

- **Commuincations and Entertainment:** telegraphy, telephony, radio broadcasting(wireless communication, TV broadcasting, satellites, tape recorders, etc.)
- **Defence applications:** RADAR (RAdio Detection And Ranging)- By using RADAR it is possible not only to detect, but also to find the exact location of the enemy aircraft.
  Guided missiles are completely controlled by by electronic circuits.
- **Industrial Applications:** Electronic circuits are used in industrial applications like control of thickness, quality weight and moisture content of a material.
  Electronic amplifier circuits are used to amplify signals and thus control the operations of automatic door-openers , lightening systems, power systems and safety devices, etc.

Applications of Electronics

- **Medical Sciences:** Doctors and scientists are constantly finding new uses of electronic systems in the diagnosis and treatment of various diseases.
  For eg., X-rays, ECG etc.
- **Intrumentation:** Instrumentation plays a very important role in any industry and research organisation, for precise measurement of various quantities.
  For eg., catohre ray oscilloscope, frequency counters, signal generators, etc.

Electrical Properties of Solids

- **Conductors**

  • e.g. copper or aluminium

  • have a cloud of free electrons (at all temperatures above absolute zero). If an electric field is applied electrons will flow causing an electric current

- **Insulators**

  • e.g. polythene

  • electrons are tightly bound to atoms so few can break free to conduct electricity

Electrical Properties of Solids

- **Semiconductors**

  • e.g. silicon (Si), germanium (Ge), gallium arsanide (GaAs)

  • at very low temperatures these have the properties of insulators

  • as the material warms up some electrons break free and can move about, and it takes on the properties of a conductor - albeit a poor one

  • however, semiconductors have several properties that make them distinct from conductors and insulators

Semiconductors

- **Pure or Intrinsic Semiconductors**
  • thermal vibration results in some bonds being broken generating **free electrons** which move about
  • these leave behind **holes** which accept electrons from adjacent atoms and therefore also move about
  • electrons are **negative charge carriers**
  • holes are **positive charge carriers**
- At room temperatures there are few charge carriers
  • pure semiconductors are poor conductors

- **Extrinsic Semiconductors**

  • The characteristics of a semiconductor material can be altered significantly by the addition of specific impurity atoms to the relatively pure smiconductor material.

  • A semiconductor material that has been subjected to the doping process is called an **Extrinsic material**.

  • There are two extrinsic materials used for semiconductor device fabrication: **p-type and n-type**.

n-type material

- An n-type material is created by introducing impurity elements that have five valence electrons (pentavalent), such as antimony, arsenic and phosporus. These dopnts are Group V elements.
- The diagram shows the extra electron that will be present when a Group V dopant is introduced to a material such as silicon.

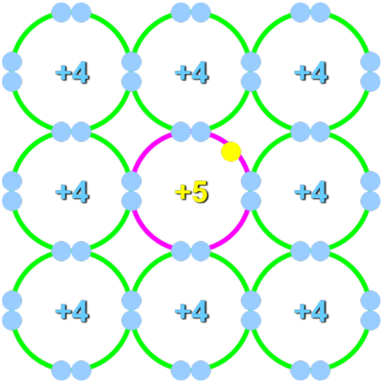


Figure 1

n-type material

- There is an additional fifth electron due to the impurity atom, which is unassociated with any particular covalent bond.
- This remaining electron loosely bound to its particular parent (antimony) atom, is relativley free to move within the newly formed n-type material.
- Diffused impurities with five valence electrons (pentavalent) are called **donor atoms**, since they have donated a relatively "free" electron to the structure.

p-type material

- An p-type material is created by introducing impurity elements that have three valence electrons (trivalent), such as boron, gallium, indium, and aluminium. These dopnts are Group III elements.
- The diagram shows the hole that will be present when a Group III dopant is introduced to a material such as silicon.

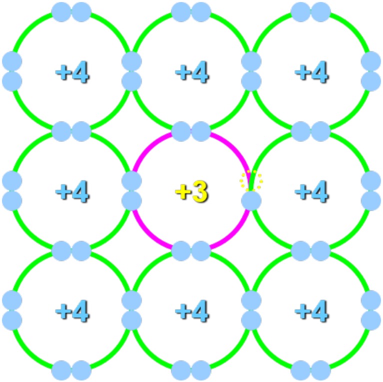


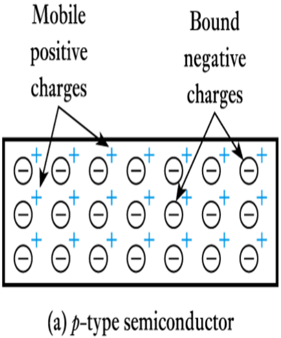
Figure 2

p-type material

- There is an insufficient number of electrons to complete the covalent bonds of the newly formed lattice.
- The resulting vacancy is called hole.
- Diffused impurities with three valence electrons(trivalent) are called **acceptor atoms**, since the resulting vacancy will readily accept a free electron.

Majority and Minority Carriers

- In an n-type material, the electrons are called the **majority charge carriers** and the holes the **minority charge carriers**
- In an p-type material, the holes are called the **majority charge carriers** and the electrons are the **minority charge carriers**



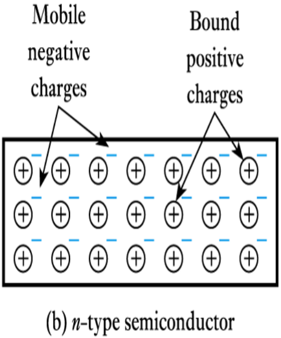


Figure 3

- **Steady State:** When no external source is connected to the pn junction, diffusion and drift balance each other out for both the holes and electrons
- **Space Charge Region:** Also called the depletion region. This region includes the net positively and negatively charged regions. The space charge region does not have any free carriers. The width of the space charge region is denoted by W in pn junction formula's.
- **Metallurgical Junction:** The interface where the p- and n-type materials meet.
- It is able to conduct current in one direction only and in the reverse direction, it offers very high resistance.

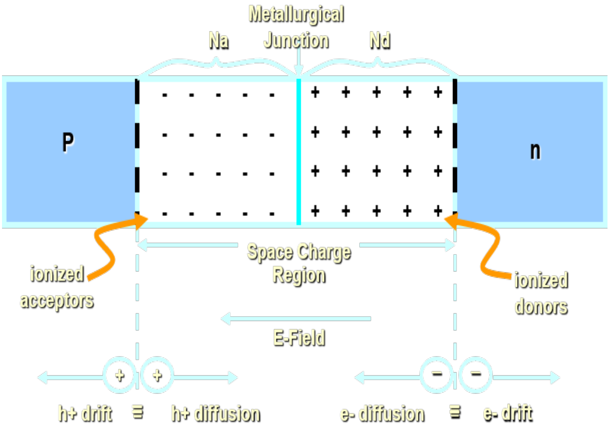
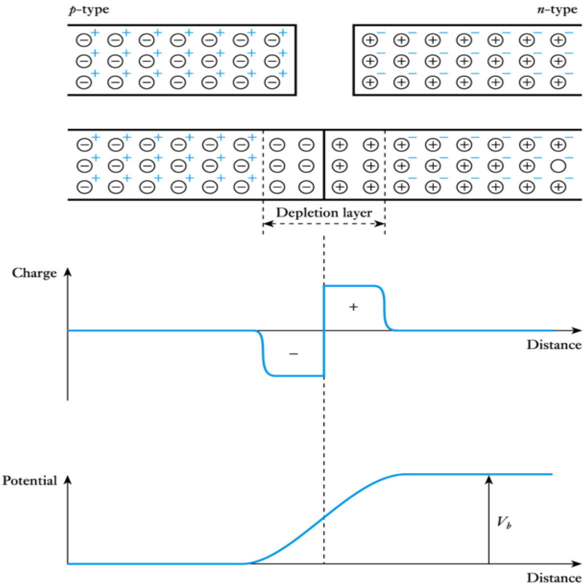

Metallurgical
Na
Junction
Nd
P
n
ionized
acceptors
Space Charge
Region
ionized
donors
E-Field
h+ drift
≡
h+ diffusion
e- diffusion
≡
e- drift

Figure 4

- When p-type and n-type materials are joined this forms a **pn junction**.
- majority charge carriers on each side diffuse across the junction where they combine with charge carriers of the opposite polarity
- holes from p region diffuses into the n region. They then combines with the free electrons in the n region.
- electrons from n region diffuses into the p region. They then combines with the free holes in the p region.
- this diffusion takes place because they move due to thermal energy and also due to difference in their concentrations in the two regions.

- The holes from p-region and electrons from n region moves towards each other and combine.
- Thus all the holes and electrons would have been eliminated.
- But actually the diffusion of electrons and holes acroos the junction occurs fro a very short time.
- After a few recombinations of electrons and holes in the neighbourhood of the junction, a restraining force called barrier is set up automatically.
- Further diffusion of electrons and holes from one side to other is stopped by this barrier.

- Some of the holes in the p region and some of the free electrons in the n region diffuse towards each other and recombine.
- Each recombination eliminates a hole and a free electron.
- The negative acceptor ions in the p region and positive donor ions in the n region in the neighbourhood of the junction are left uncompensated.
- Additional holes trying to diffuse into the n region are repelled by the uncompensated positive charge of the donor ions and vice versa.
- As a result, total recombination of electrons and holes cannot occur.



p-type
n-type
Depletion layer
Charge
+
–
Distance
Potential
V_b
Distance

Figure 5

- The region containing the uncompensated acceptor and donor ions is called **depletion region**. This region is depleted of mobile charges.
- Since this region has immobile (fixed) ions which are electrically charged, it is also called **space charge region**.
- The electric field between the acceptor and donor ions is called **barrier**.
- **Barrier potential** is the difference of the potential from one side of the barrier to the other side of the barrier. This barrier discourages the diffusion of majority carriers across the juction.
- For Si and Ge the barrier potential is approximately 0.7V and 0.3V respectively.

- What happens to minority carriers?
- The barrier helps the minority carriers to drift across the junction.
- The minority carriers are constantly generated due to thermal energy. But these are not responsible for the generation of the current.
- Electric current cannot flow since no circuit has been connected to the PN junction.
- The drift of minority carriers across the junction is counterbalanced by the diffusion of the same number of majority carriers across the junction.

The Biased PN Junction Diode

- The pn junction is considered biased when an external voltage is applied.
- There are two types of biasing: Forward bias and Reverse bias.

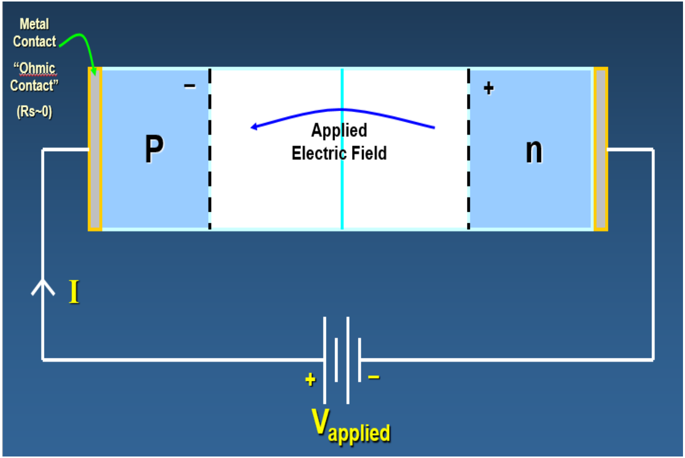


Figure 6

Forward Bias

- If the p side is connected to the positive terminal of the battery and the n side is connected to the negative terminal of the battery, the PN junction is said to be forward biased.
- The holes are repelled from the positive terminal of the battery and are compelled to move towards the junction.
- The elctrons are repelled from the negative terminal of the battery and drift towards the junction.
- Because of the acquired energy, some of the holes and free electrons penetrate the depletion region. This reduces the potential barrier.
- The width of the depletion region reduces and so does the barrier height.
- As a result, more majority carriers diffuse across the junction.

Forward Bias

- These carriers recombine and cause movement of charge carriers in the depletion region.
- For each recombination of free electrons and hole that occurs, an electron from the negative terminal of the battery enters the n type material and drift towards the junction.
- Similarly, in the p type material near the positive terminal of the battery, an electron brekas a bond in the crystal and enters the positive terminal of the battery.
- For each electron that breaks its bond, a hole is created and it drifts towards the junction.
- There is continuous electron current in the external circuit.
- The current in the p type material is due to the movement of holes and vice versa.

Forward Bias

- The current continues as long as the battery is in the circuit.
- If the battery voltage is increased, the barrier potential is further reduced.
- More majority carriers diffuse across the junction.
- This results in an increased current through the pn junction.

Reverse Bias

- If the n side is connected to the positive terminal of the battery and the p side is connected to the negative terminal of the battery, the PN junction is said to be reverse biased.
- The holes in the p region are attracted towards the negative terminal of the battery and the electrons in the n region are attracted towards the positive terminal of the battery.
- Thus the majority carriers are swept away from the junction.
- This widens the depletion region and increases the barrier potential.
- Due to increased barrier potential, the majority carriers are unable to diffuse across the junction.
- But this barrier potential helps the minority carriers to cross the junction.

Reverse Bias

- The rate of generation of minority carriers depends upon the temperature.
- If the temperature is fixed, the rate of generation of minority carriers remains constant.
- Therefore, the current due to flow of the minority carriers remains the same whether the battery voltage is low or high. This current is called **Reverse saturation current**
- This current is usually very small as the number of minority carriers is small.
- It is of the order of nanoamperes in silicon diodess and microamperes in germanium diodes.

Transition or Depletion Capacitance

- The reverse biased pn junction diode has a region of high resistivity (space charge or depletion region).
- The p and n regions acts as the plates of a capcitor and the space charge region acts as the dielectric.
- Thus the pn junction in reverse bias has an effective capacitance called the transition or depletion capacitance.

Currents in a PN Junction

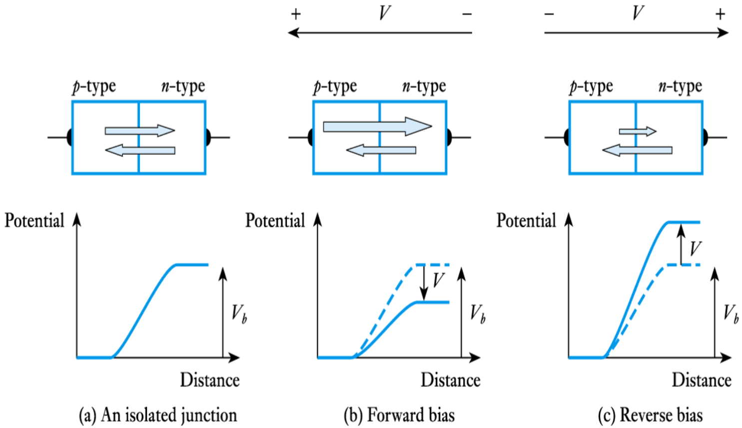


Figure 7

Reverse Breakdown

- In reverse bias, a very small current due to minority carriers flow that is independent of the voltage applied.
- But if the reverse bias is made too high, the current through the pn junction increases abruptly.
- The voltage at which this phenomenon occurs, is called **breakdown voltage**. At this voltage, the crystal structure breaks down.
- There are two processes which can cause junction breakdown.
  - **Zener Breakdown**
  - **Avalanche Breakdown**

Zener Breakdown

- When the reverse bias is increased, the electric field at the junction also increases.
- High electric field causes covalent bonds to break.
- Thus a large number of carriers are generated. This cause a large current to flow. This mechanism of breakdown is called **Zener breakdown**.

Avalanche Breakdown

- In this mechanism, the increased electric field causes increase in the velocities of minority carriers.
- These high energy carriers break covalent bonds, thereby generating more carriers.
- Again, these generated carriers are accelerated by the electric field.
- They break more covalent bonds during their travel.
- A chain reactino is thus established, created a large number of carriers.
- This gives rise to a high reverse current.
- This mechanism of breakdown mechanism is called **Avalance breakdown**.

PN junction diode symbol

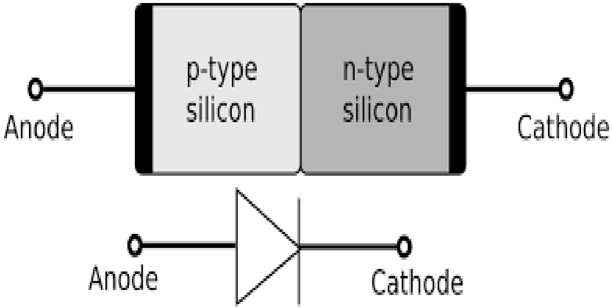


Figure 8

PN junction diode current equation

- pn junction current is given approximately by Shockley's equation:

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right) \tag{1}$$

• where I_S is the reverse saturation current, V_D is the applied voltage, η (Greek letter eta) is a constant in the range 1 to 2 determined by the junction material and V_T is the thermal voltage.
• for most purposes we can assume $\eta = 1$

$$V_T = \frac{kT}{q} \tag{2}$$

• q is the electronic charge = 1.6×10^{-19} C, k is Boltzmann's constant = 1.38×10^{-23} J/K, T is the absolute temperature in kelvins = 273 + temp in $^\circ$C

For positive values of V_D the first term of the above equation will grow very quickly. The second term can be ignored.

$$I_D \approxeq I_S e^{\frac{V_D}{\eta V_T}} \tag{3}$$

For negative values of V_D, the exponential term drops very quickly below the level of I_S, Thus

$$I_D \approxeq -I_S \tag{4}$$

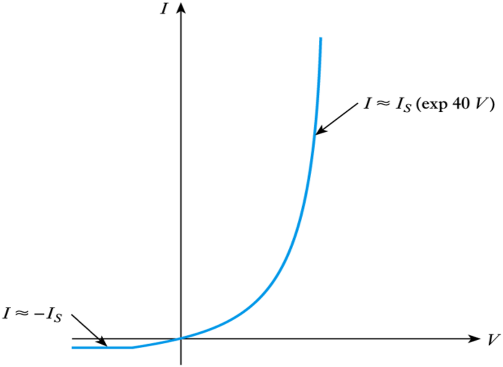


Figure 9

- Forward and reverse currents

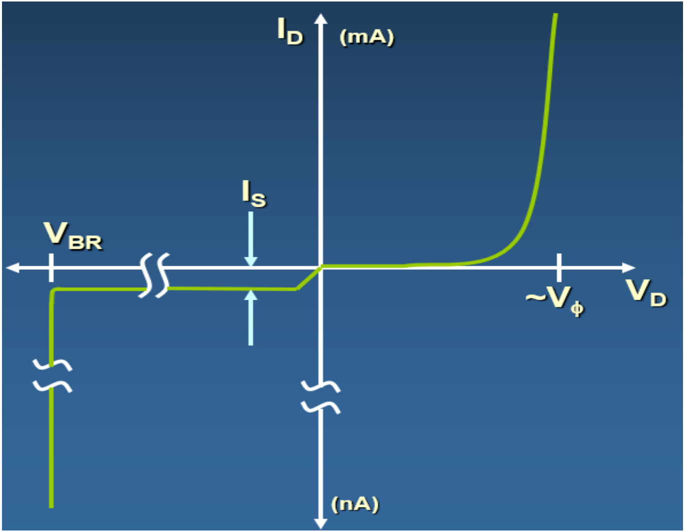


Figure 10

- Turn-on and breakdown voltages for a silicon device

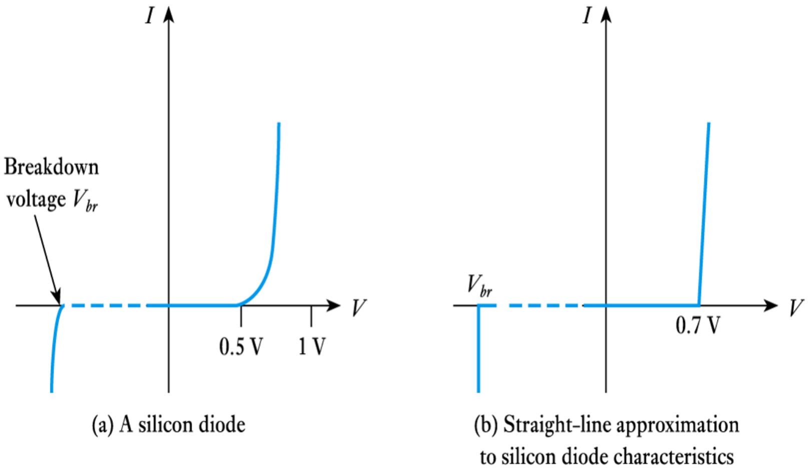
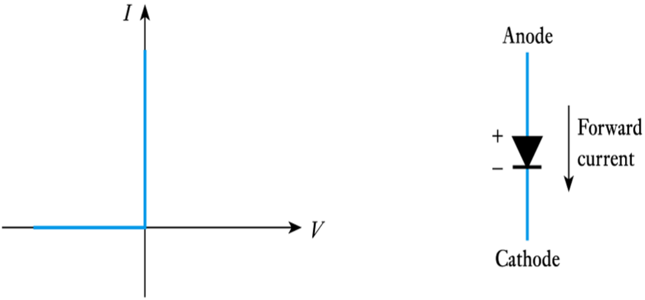


Figure 11

Diodes

- An ideal diode passing electricity in one direction but not the other.



(a) I–V characteristic (b) Diode circuit symbol

Figure 12

Zener region

- In the negative region, there is a point where the application of two negative voltage will result in a sharp change in the characteristic.
- The current increases at a very rapid rate in a direction opposite to that of the positive voltage region.
- The reverse bias potential that results in this change in characteristics is called zener potential and is given by the symbol V_z.
- The maximum reverse bias potential that can be applied before entering the zener region is called the peak inverse voltage or the peak reverse voltage.

Zener region

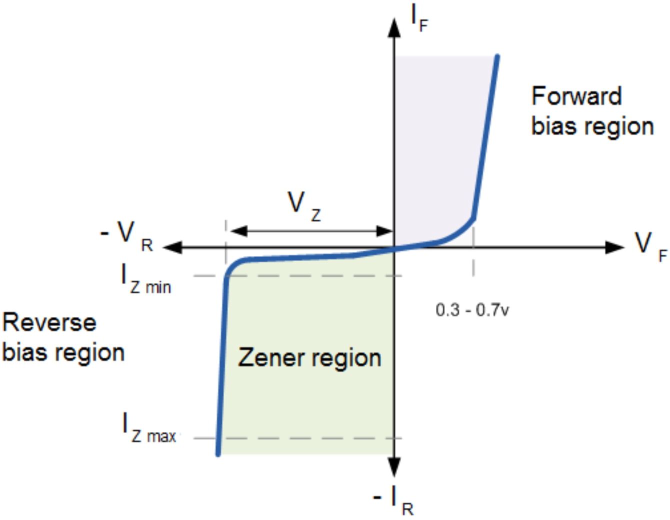
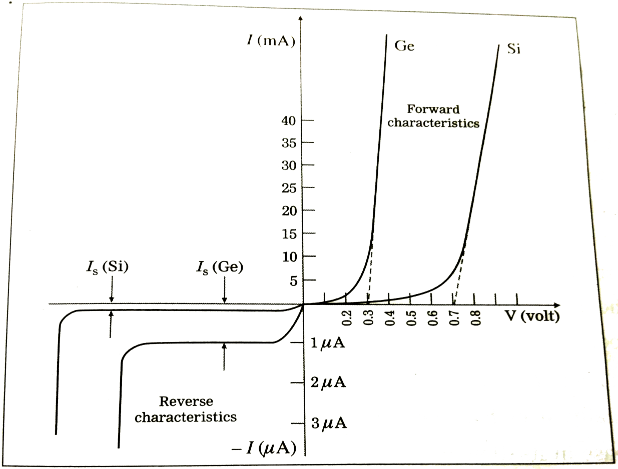


Figure 13

Comparison of Si and Ge



Junction diode characteristics, Ge and Si

Figure 14

Temperature Effects

- In the forward bias region the characteristics of a silicon diode shift to the left at a rate of 2.5 mV per centigrade degree increase in temperature.

$$\frac{dV}{dT} \approx -2.5mV/^{\circ}C \tag{5}$$

- In the reverse bias region, the reverse saturation current of a silicon diode doubles for every $10^{\circ}C$ rise in temperature.
- If I=I_{01}, at T=T_1, then at a temperature T, I_0 is given by

$$I_0(T) = I_{01}x2^{(T-T_1)/10} \tag{6}$$

Comparison of Si and Ge

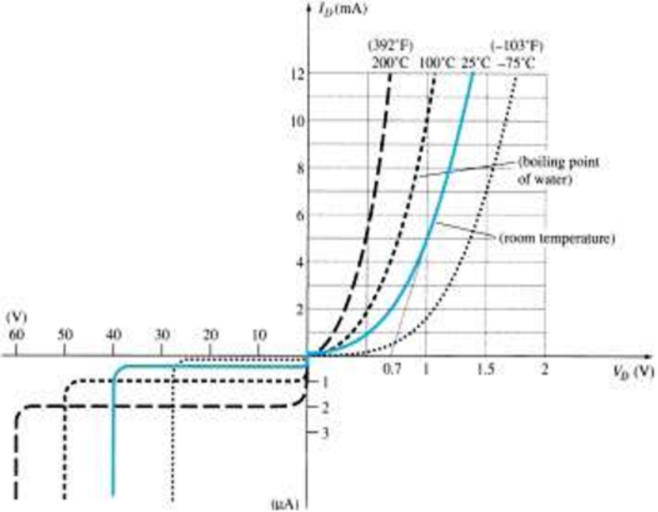


Figure 15

Ideal Diode

- The semiconductor diode behaves like a mechanical switch thus it can control whether current will flow between its two terminals.

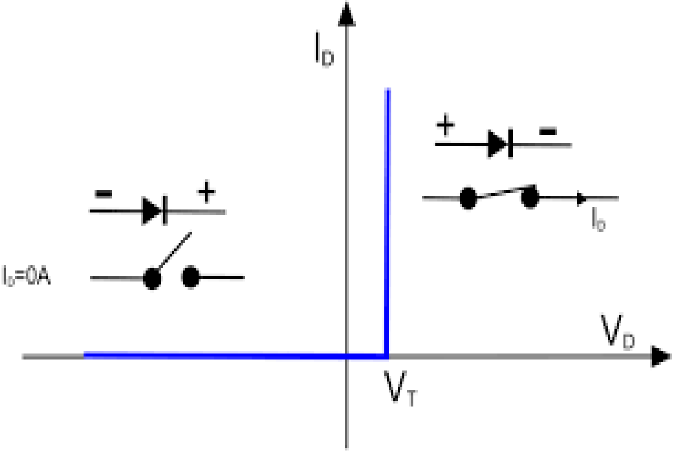


Figure 16

Ideal Diode

- Ideally if the semiconductor diode is to behave like a closed switch in the forward bias region, the resistance of the diode should be 0 Ω.
- In the reverse bias region, its resistance should be ∞ Ω to represent the open circuit equivalnet.

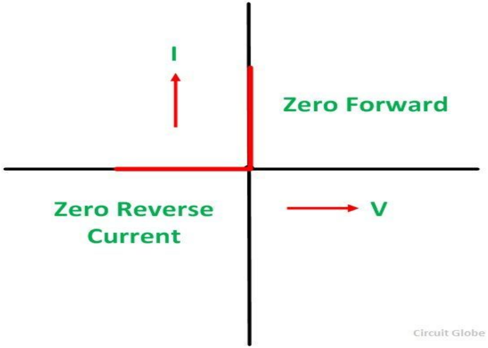


Figure 17

Diode Resistance

- **DC or Static Resistance**
  • When a dc voltage is applied to a circuit containing diode, it will result in an operating point on the characteristics curve that will not change with time.

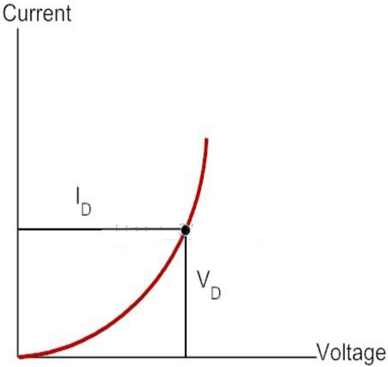


Figure 18

Diode Resistance

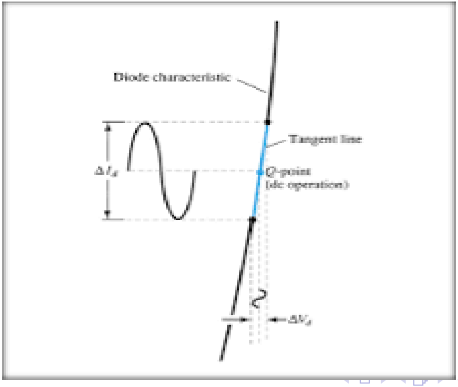
- The dc resistance of the diode at the operating point is given by

$$R_D = \frac{V_D}{I_D} \tag{7}$$

- The higher the current through a diode, the lower is the dc resistance value.

Diode Resistance

- **AC or Dynamic Resistance**
  • When a sinusoidal input is applied, the varying input will move the operating point up and down on the characteristics and thus defines a specific change in current and voltage.
  • With no applied voltage, the point of operation would be the Q-point (Quiescent point means "still or unvarying").



Diode Resistance

- The ac resistance of the diode at the operating point is given by

$$r_d = \frac{\Delta V_d}{\Delta I_d} \tag{8}$$

- The lower the Q point of operation, the higher is the ac resistance.

Diode Resistance

$$I_D = I_S\left(e^{\frac{V_D}{\eta V_T}} - 1\right) \tag{9}$$

$$\frac{dI_D}{dV_D} = I_S\frac{1}{\eta V_T}\left(e^{\frac{V_D}{\eta V_T}}\right) = \frac{1}{\eta V_T}(I_D + I_S) \tag{10}$$

In general, $I_D >> I_S$

$$\frac{dI_D}{dV_D} \approx \frac{I_D}{\eta V_T} \tag{11}$$

$$\frac{dV_D}{dI_D} = r_d = \frac{\eta V_T}{I_D} \tag{12}$$

Subsituting $\eta = 1$ and $V_T \approx 26$ mV

$$r_d = \frac{26mV}{I_D} \tag{13}$$

Diode Equivalent Circuits

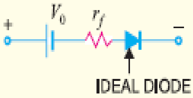
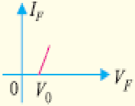
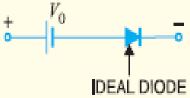
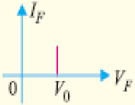
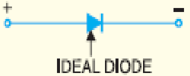
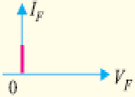
S.No.	Type	Model	Characteristic
1.	Approximate model		
2.	Simplified model		
3.	Ideal Model		

Figure 20

- **Diffusion Capacitance**
  • This capacitance is due to the injected charge stored near the junction outside the depletion region.
  • It is defined as the rate of change of injected charge with voltage, called the diffusion or storage capacitance, C_D.

$$C_D = \frac{dQ}{dV} = \tau \frac{dI}{dV} = \tau g = \frac{\tau}{r} = \frac{\tau I}{\eta V_T} \tag{14}$$

- **Transistion or Space-charge or Depletion Capacitance**
  • When a pn junction is reverse biased, the depletion region acts like an insulator or as a dieletric material. The p and n regions act as plates. Thus creating a capacitance called transition capacitance.

$$C_T = \frac{\epsilon A}{d} \tag{15}$$

  • This is used in a device called varicap or varacter.

Diode Capacitances

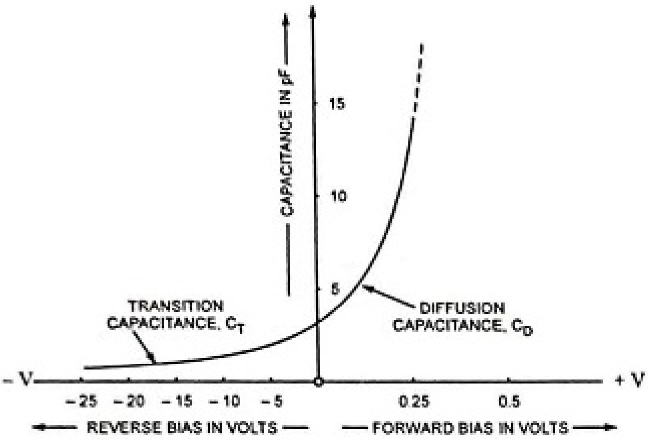


Figure 21

Zener Diodes

- Diode in the zener region is used in the design of Zener diode.
- For the zener diode the direction of conduction is opposite to that of the arrow in the symbol.

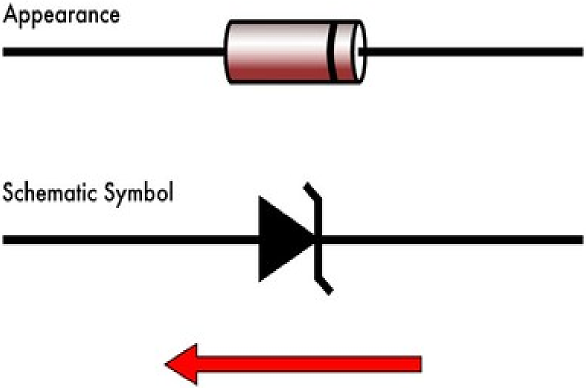


Figure 22

Zener Diodes

- The location of the zener region can be controlled by varying the doping levels.
- An increase in doping that produces an increase in the number of added impurities, will decrease the Zener potential.
- The zener potential of a zener diode is very sensitive to the temperature of operation.
- The temperature coefficient can be used to find the change in zener potential due to a change in temperature using the equation:

$$T_C = \frac{\frac{\Delta V_Z}{V_Z}}{T_1 - T_0} x100\%/^0C \tag{16}$$

where T_1 is the new temperature level, T_0 is the room temperature, T_C is the temperature coefficient and V_Z is the nominal zener potential at 25^0C.

Zener diode as a voltage regulator

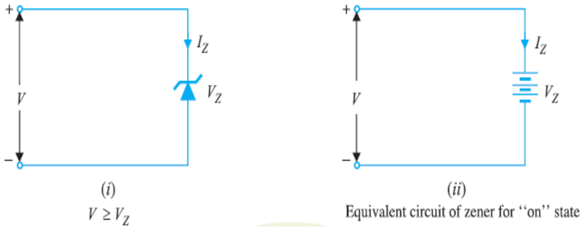


Figure 23

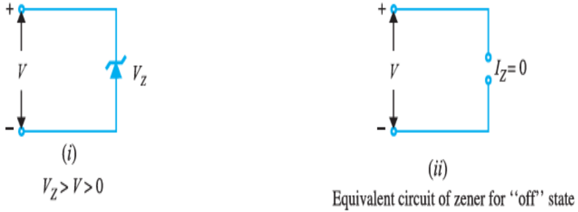


Figure 24

Zener diode as a voltage regulator

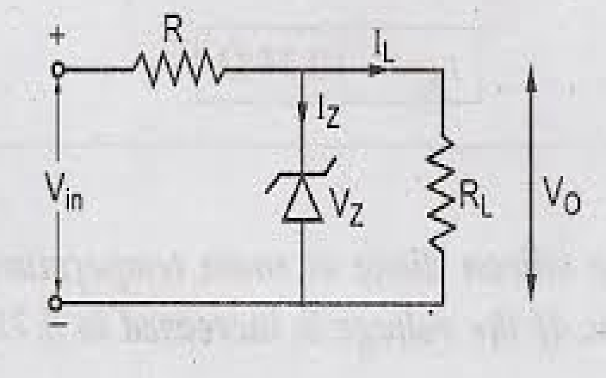


Figure 25

Case 1: When V_i and R are fixed.

- Determine the state of the zener diode by removing it from the network and calculating the voltage across the resulting open circuit.

$$V = V_L = \frac{R_L V_i}{R + R_L} \tag{17}$$

- If V $\geq V_Z$, zener diode is on and if V $\leq V_Z$, the zener diode is off.

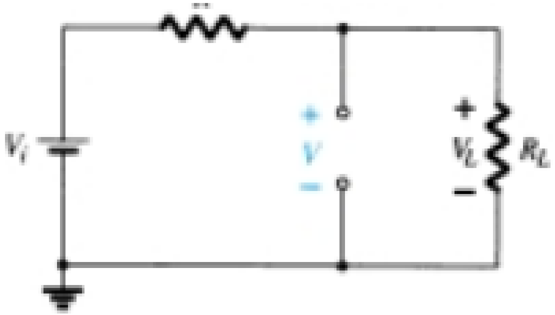


Figure 26

- If it is in on state:

$$V_L = V_Z \tag{18}$$

$$I_R = I_Z + I_L => I_Z = I_R - I_L \tag{19}$$

$$P_Z = V_Z I_Z \tag{20}$$

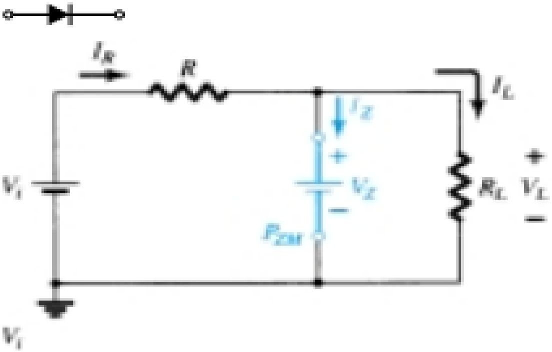


Figure 27

Ex. (a)Determine V_L, V_R, I_Z, P_Z (b) Repeat with $R_L = 3\ \text{k}\Omega$.

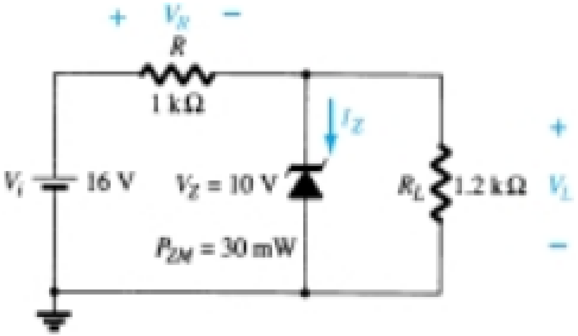


Figure 28

$$V = \frac{R_L}{R + R_L} V_i = 8.73V \quad (21)$$

Since V=8.73 V is less than $V_Z = 10$ V, diode is in off state.
(a) V_L=V=8.73 V; V_R=$V_i - V_L$ = 16-8.73 = 7.27 V
I_Z=0; $P_Z = V_Z I_Z = 0W$

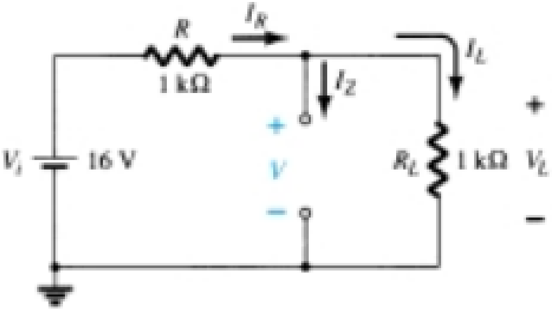


Figure 29

(b)

$$V = \frac{R_L}{R + R_L} V_i = 12V \tag{22}$$

Since V=12 V is greater than $V_Z = 10$V, diode is in on state.

(a) $V_L = V_Z$=10 V; V_R=$V_i - V_L$ = 16-10 = 6 V

I_L =V_L/R_L = 10 V/ 3 $k\Omega$ = 3.33 mA & I_R =V_R/R = 6 V/ 1 $k\Omega$ =6 mA

$P_Z = V_Z I_Z = 10x2.67 = 26.7 mW$

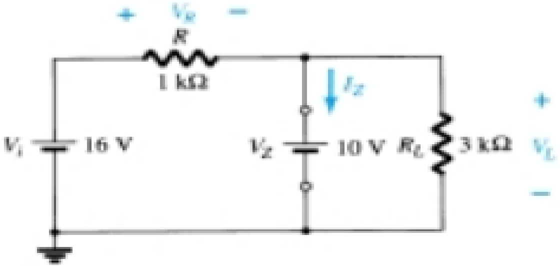


Figure 30

Case 2: Fixed V_i and Variable R_L

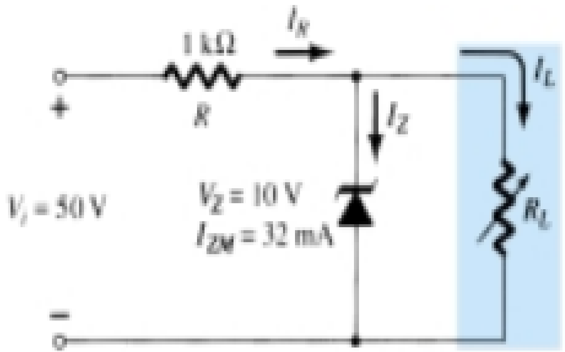


Figure 31

- Due to V_Z, there is a specific range of resistor values and I_L, which will ensure that the Zener is in "on" state.
- If R_L is too small, it will result in a voltage V_L less than V_Z and Zener diode is in "off" state.
- Minimum load resistance that will turn the Zener diode "on" will be that results in load voltage $V_L = V_Z$.

$$V_L = V_Z = \frac{R_L V_i}{R + R_L} \tag{23}$$

$$R_{L_{min}} = \frac{R V_Z}{V_i - V_Z} \tag{24}$$

- Load resistance greater than $R_{L_{min}}$ will ensure that the Zener diode is in the "on" state.

$$I_{L_{max}} = \frac{V_L}{R_{L_{min}}} = \frac{V_Z}{R_{L_{min}}} \tag{25}$$

$$V_R = V_i - V_Z \tag{26}$$

$$I_R = \frac{V_R}{R} \tag{27}$$

$$I_Z = I_R - I_L \tag{28}$$

$$I_{L_{min}} = I_R - I_{ZM} \tag{29}$$

$$R_{L_{max}} = \frac{V_Z}{I_{L_{min}}} \tag{30}$$

(a)Find the range of R_L and I_L that will result in V_{R_L} being maintained at 10 V. (b) Determine P_{max}

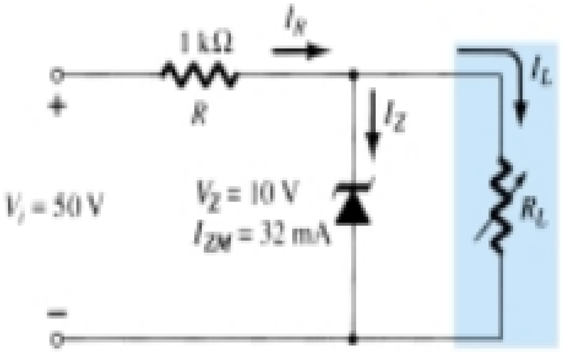


Figure 32

- Ans: $R_{L_{min}}$=250Ω; V_R=40 V; I_R=40 mA; $I_{L_{min}}$=8 mA; $R_{L_{max}}$=1.25 kΩ; P_{max}=320 mW

• For fixed values of R_L, the voltage V_i must be sufficiently large to turn the Zener didoe "on".

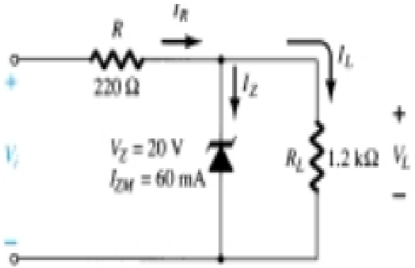


Figure 33

$$V_L = V_Z = \frac{R_L V_i}{R + R_L} \tag{31}$$

$$V_{i_{min}} = \frac{(R_L + R)V_Z}{R_L} \tag{32}$$

- The maximum value of V_i is limited by the maximum Zener current I_{ZM}.

$$I_{R_{max}} = I_{ZM} + I_L \tag{33}$$

$$V_{i_{max}} = V_{R_{max}} + V_Z \tag{34}$$

$$V_{i_{max}} = I_{R_{max}} R + V_Z \tag{35}$$

- Determine the range of values of V_i that will maintain the Zener diode in the on state.

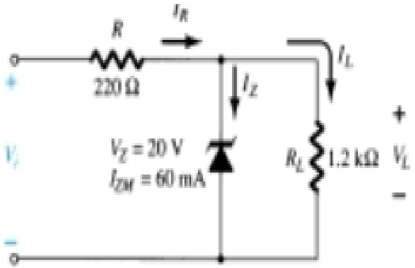


Figure 34

- Ans: V_{imin}= 23.67 V; I_L= 16.67 mA; $I_{R_{max}}$=76.67 mA; $V_{i_{max}}$= 36.87 V

Load Line Analysis

• Solving the circuit is all about finding the current and voltage levels that will satidfy both the characteristics of the diode and the chosen network parameters at the same time.

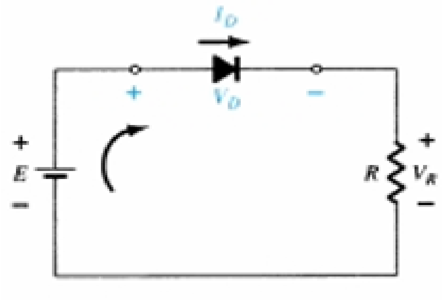


Figure 35

- The straight line is called a **load line** because the intersection on the vertical axis is defined by the applied load R. This analysis is called load line analysis.
- The intersection of the two curves will define the solution for the network and define the current and voltage levels for the network.

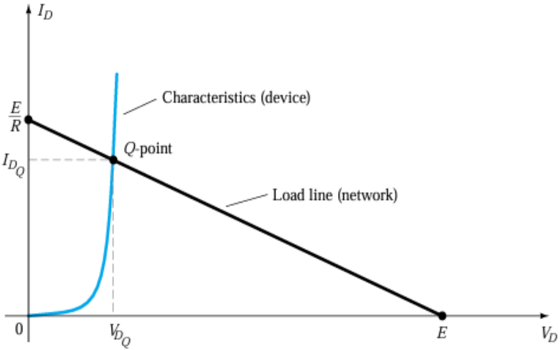


Figure 36

- In the circuit, applying Kirchoff's voltage law in the clockwise direction, we get:

$$+E - V_D - V_R = 0 => E = V_D + I_D R \tag{36}$$

- Load line can be drawn as:
  - If V_D=0 V,

$$E = V_D + I_D R = 0V + I_D R \tag{37}$$

$$I_D = \frac{E}{R}\Big|_{V_D=0V} \tag{38}$$

  - If I_D=0 A,

$$E = V_D + I_D R = V_D + (0A)R \tag{39}$$

$$V_D = E\Big|_{I_D=0A} \tag{40}$$

- A straight line drawn between these two points will define the load line.

Q.1 For the series diode configuration of the given figure, determine V_D, V_R, and I_D

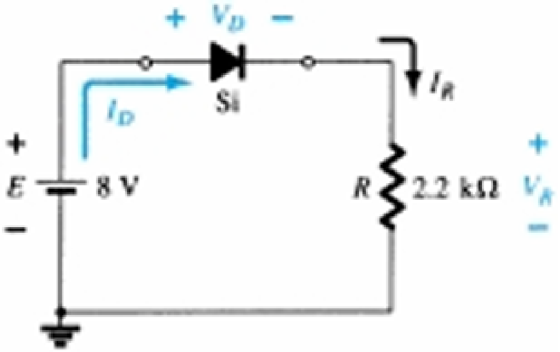


Figure 37

Q.2 Repeat with the diode reversed.

Q.3 For the series diode configuration of the given figure, determine V_D, V_R, and I_D

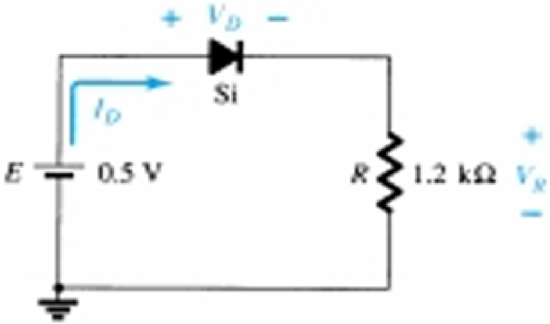


Figure 38

Q.4 For the series diode configuration of the given figure, determine V_0 and I_D

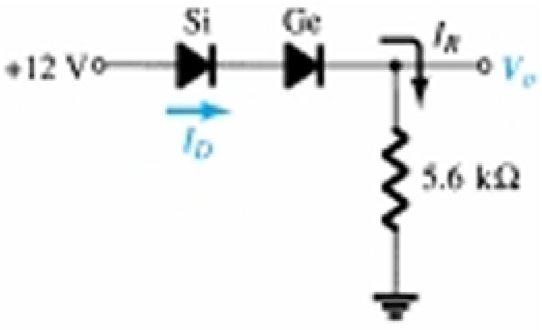


Figure 39

Q.5 For the series diode configuration of the given figure, determine V_{D2}, V_0 and I_D

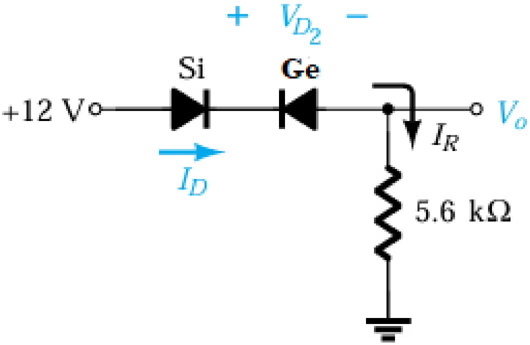


Figure 40

Q.6 For the parallel diode configuration of the given figure, determine V_0, I_1, I_{D1} and I_{D2}.

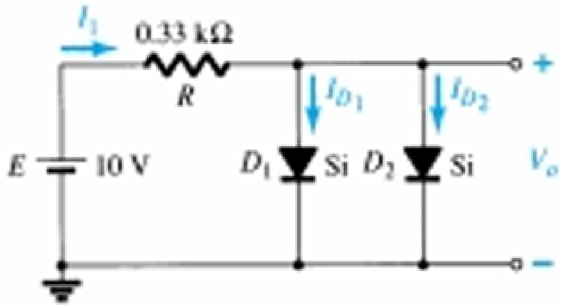


Figure 41

Ans: V_0=0.7 V; I_1=28.18 mA; I_{D1}=I_{D2}=14.09 mA

Q.7 Determine the currents I_1, I_2 and I_{D2} for the following network.

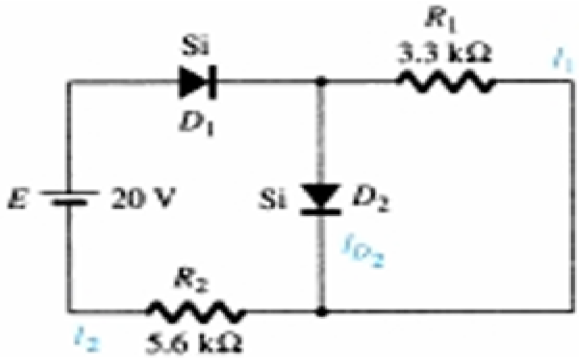


Figure 42

Hint for Q.7

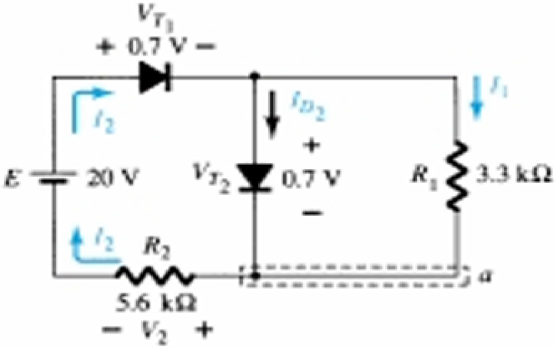


Figure 43

Ans: I_1=0.212 mA; I_2=3.32 mA; $I_{D2} = 3.11mA$

RECTIFIERS

- Rectifier is a device which converts the sinusoidal ac voltage into either positive or negative pulsating dc.
- PN junction diode, which conducts in forward bias and does not conduct in reverse bias, can be used for rectification.
- Rectifiers needs one, two or four diodes.
- The output voltage from a rectifier circuit has a pulsating character, i.e. it contains ac components along with dc component.

Half Wave Rectifier

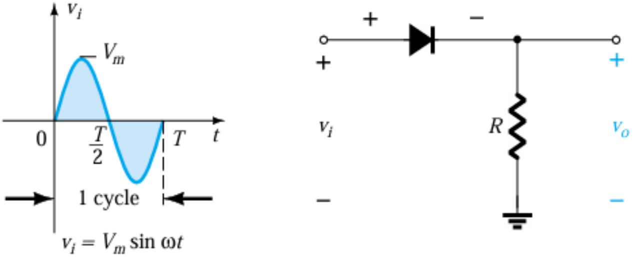


Figure 44

Conduction Period: 0——T/2

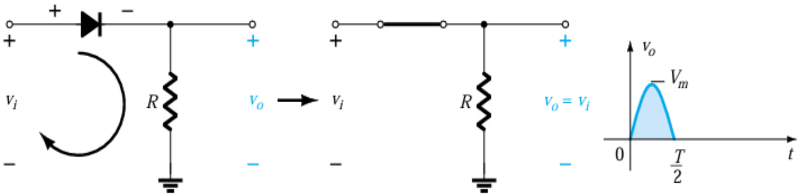


Figure 45

Non Conduction Period: T/2——T

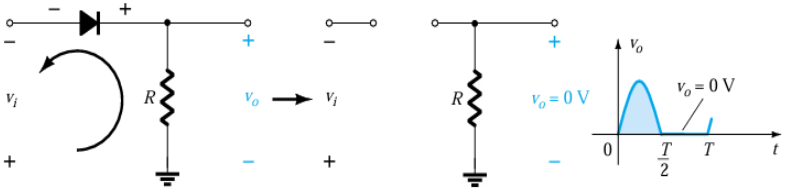


Figure 46

Output Waveform

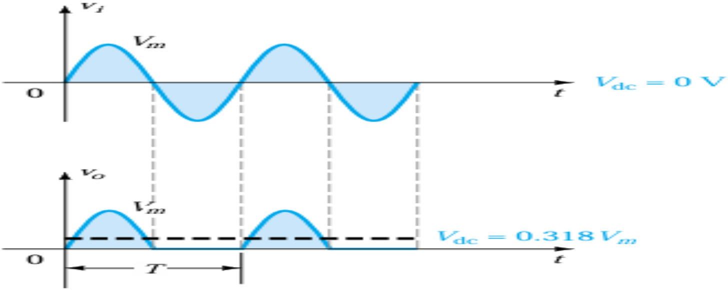


Figure 47

Silicon diode having $V_T = 0.7$ V

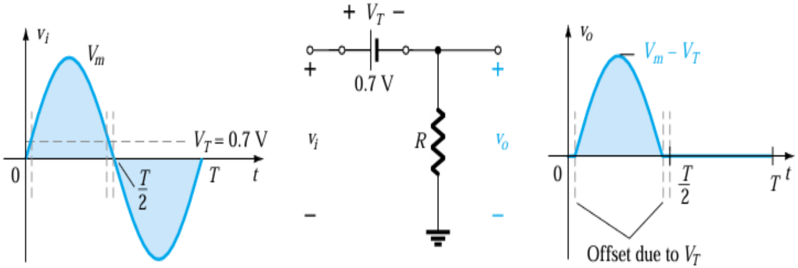


Figure 48

PIV

- It is the voltage rating that must not be exceeded in the reverse-bias region or the diode will enter the Zener avalanche region.
- $PIV_{halfwaverectifier} = V_m$

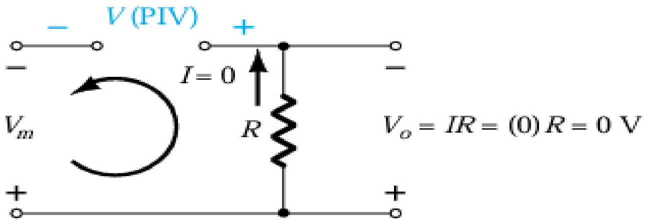


Figure 49

Peak current

- The input voltage applied to the rectifier is

$$V_i = V_{max} sin(\omega t) \tag{41}$$

$$I_{max} = \frac{V_{max}}{R_F + R_L} \tag{42}$$

DC output current

- Average or DC value of the output current is given by:

$$I_{dc} = \frac{1}{2\pi}\int_0^{2\pi} I \, d(\omega t)$$

$$I_{dc} = \frac{1}{2\pi}\left[\int_0^{\pi} I_{max} sin\omega t \, d(\omega t) + \int_{\pi}^{2\pi} 0 \, d(\omega t)\right]$$

$$I_{dc} = \frac{1}{2\pi}\left[I_{max}\{-cos\omega t\}_0^{\pi}\right] = \frac{I_{max}}{\pi} = 0.318 I_{max}$$

or $I_{dc} = \frac{V_{max}}{\pi(R_F+R_L)} = \frac{V_{max}}{\pi R_L}$ if $R_L >> R_F$

DC output voltage

- Average or DC value of the output voltage across the load is given by:

$$V_{dc} = I_{dc} R_L = \frac{V_{max}}{\pi(R_F + R_L)} R_L = \frac{V_{max}}{\pi(1 + \frac{R_F}{R_L})}$$

If $R_L >> R_F$ then

$$V_{dc} \approxeq \frac{V_{max}}{\pi}$$

RMS value of current

$$I_{rms}^2 = \frac{1}{2\pi}\int_0^{2\pi} I^2\, d(\omega t)$$

$$I_{rms} = \frac{1}{2\pi}\left[\,\int_0^{\pi} I_{max}^2 sin^2\omega t\, d(\omega t) + \int_{\pi}^{2\pi} 0\, d(\omega t)\right]$$

$$= \frac{I_{max}^2}{4}$$

$$I_{rms} = \frac{I_{max}}{2}$$

or $I_{rms} = \frac{V_{max}}{2(R_F + R_L)}$

RMS value of output voltage

$$V_{L_{rms}} = I_{rms} R_L = \frac{V_{max} R_L}{2(R_F + R_L)} = \frac{V_{max}}{2(1 + \frac{R_F}{R_L})}$$

If $R_L >> R_F$,

$$V_{L_{rms}} = \frac{V_{max}}{2}$$

Rectification Efficiency

- It is defined as the ratio of dc output power to the ac input power, $\eta = \frac{P_{dc}}{P_{ac}}$

$$P_{dc} = I_{dc}^2 R_L = (\frac{I_{max}}{\pi})^2 R_L$$

P_{ac} = Power dissipated in diode junction + power dissipated in load resistance R_L

$$I_{rms}^2 R_F + I_{rms}^2 R_L = (\frac{I_{max}}{2})^2 (R_F + R_L) = \frac{I_{max}^2}{4}(R_F + R_L)$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{(\frac{I_{max}}{\pi})^2 R_L}{\frac{I_{max}^2}{4}(R_F + R_L)} = \frac{4}{\pi^2}\frac{R_L}{R_F + R_L}$$

$$\eta = \frac{0.406}{1 + \frac{R_F}{R_L}}$$

If $R_L >> R_F$ then $\eta \approx 0.406$ or 40.6 %

Ripple Factor

- The pulsating output of a rectifier contains a dc component and ac components called the **ripples**.
- The ripple voltage or current is measured in terms of the ripple factor which is defined as the ratio of the effective value of the ac components of voltage(current) present in the output from the rectifier to the dc or average value of the output voltage(current). The effective value of the load current is given as:

  $$I_{rms}^2 = I_{dc}^2 + I_1^2 + I_2^2 + I_4^2 + - - - - - -- = I_{dc}^2 + I_{ac}^2$$
  where I_1, I_2, I_4 etc. are the rms values of fundamental, second, fourth etc. harmonics and I_{ac}^2 is the sum of the squares of the rms vlaues of the ac components.
- Ripple factor,
  $\gamma = \frac{I_{ac}}{I_{dc}} = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}} = \sqrt{(\frac{I_{rms}}{I_{dc}})^2 - 1} = \sqrt{K_f^2 - 1}$ where K_f is the form

  factor of the input voltage.

Ripple Factor

- For half-wave rectifier, form factor is given as
  $K_f = \frac{I_{rms}}{I_{dc}} = \frac{\frac{I_{max}}{2}}{\frac{I_{max}}{\pi}} = \frac{\pi}{2} = 1.57$
  Ripple factor, $\gamma = \sqrt{1.57^2 - 1} = 1.21$

Transformer Utilization Factor

- It is define as the ratio of power delivered to the load and ac rating of the transformer secondary.

$$TUF = \frac{P_{dc}}{P_{ac_{(rating)}}} = \frac{I_{dc}^2 R_L}{V_{rms} I_{rms}} = \frac{(\frac{I_{max}}{\pi})^2 R_L}{\frac{V_{max}}{\sqrt{2}} \frac{I_{max}}{2}} = \frac{2\sqrt{2}}{\pi^2} . \frac{I_{max} R_L}{V_{max}}$$

$$V_{max} = I_{max}(R_F + R_L)$$

$$TUF = \frac{2\sqrt{2}}{\pi^2} . \frac{I_{max} R_L}{I_{max}(R_F + R_L)} = \frac{0.286 R_L}{R_L + R_F}$$

- If $R_L >> R_F$ then **TUF=0.286**.

Advantages and Disadvantages

- **Advantage:**

  • Simple circuit and low cost.

- **Disadvantages:**

  • The output current in the load contains ac components along with dc components. Ripple factor is high and extensive filtering is required to give steady dc output.

  • The power output thus rectification efficiency is quite low because power is delivered only half the time.

  • TUF is low.

Numericals

- Q.1 The turns ratio of a transformer used in a half wave rectifier is n_1:n_2=12:1. The primary is connected to the power mains: 220 V, 50 Hz. Assuming the diode resistance in forward bias to be zero, calculate the dc voltage acroos the load. What is the PIV of the diode?

  Ans: V_P=311 V; V_m=25.9 V; V_{dc}=8.24 V; PIV=25.9 V

- Q.2 A half wave rectifier uses a diode with an equivalent forward resistance of 0.3 Ω. If the input ac voltage is 10 V (rms) and the load is a resistance of 2.0 Ω, Calculate I_{dc} and I_{rms} in the load.

  Ans: I_{max}=6.15 A; I_{dc}=1.958 A; I_{rms}=3.075 A

Numericals

- Q.3 A half wave rectifier uses a diode with a forward resistance of 100 Ω. If the input ac voltage is 220 V (rms) and the load resistance is of 2 KΩ, determine (i) I_{max}, I_{dc} and I_{rms} (ii) peak inverse voltage when the diode is ideal (iii) load output voltage (iv) dc output power and ac input power (v) ripple factor (vi) TUF and (vi) rectification efficiency.

  Ans: I_{max}=148.156 mA, I_{dc}=47.16 mA and I_{rms}=74.078 mA; PIV = 34.127 V; V_{dc}= 94.32 V; P_{dc}=4.448 W; P_{ac}=11.524 W; Ripple factor(γ)= 1.21; TUF=0.2724; efficiency(η)= 38.6 %

Solution of Numerical 1

- The max. primary voltage is
  $V_P = \sqrt{2}V_{rms} = \sqrt{2}x220 = 311V$
- The max. secondary voltage is
  $V_m = \frac{n_2}{n_1}V_P = \frac{1}{12}x311 = 25.9V$
- The dc load voltage is
  $V_{dc} = \frac{V_m}{\pi} = \frac{25.9}{\pi} = 8.24V \;\; PIV = V_m = 25.9V$

Solution of Numerical 2

- V_{rms}= 10 V
- V_m= $10\sqrt{2}$ V
- $R_F = 0.3\ \Omega$
- $R_L = 2\ \Omega$
- Peak value of current in load,
  $I_{max} = \frac{V_{max}}{R_F+R_L} = \frac{10\sqrt{2}}{2+0.3} = 6.15A$
- DC output current
  $I_{dc} = \frac{I_{max}}{\pi} = \frac{6.15}{\pi} = 1.958A$
- RMS value of output current
  $I_{rms} = \frac{I_{max}}{2} = 3.075A$

Solution of Numerical 3

- V_{rms}= 220 V
- V_m= $220\sqrt{2}$ V
- R_F = 100 Ω 0.1K Ω
- R_L = 2K Ω

(i) $I_{max} = \frac{V_{max}}{R_F+R_L} = \frac{220\sqrt{2}}{2+0.1}x10^3 = 148.56mA$

$I_{dc} = \frac{I_{max}}{\pi} = \frac{148.56}{\pi} = 47.16mA$

$I_{rms} = \frac{I_{max}}{2} = 74.078mA$

(ii) $PIV = V_{max} = 220\sqrt{2} = 311.127V$

(iii) $V_{dc} = I_{dc}R_L = 47.16x10^{-3}x2x10^3 = 94.32V$

(iv) $P_{dc} = I_{dc}^2 R_L = 4.448W$

(v)$P_{ac} = \frac{I_{max}^2}{4} R_F + R_L = 11.524W$

(vi) Ripple Facotr, $\gamma = 1.21$

(vii) $TUF = \frac{0.286}{1+\frac{R_F}{R_L}} = 0.2724$

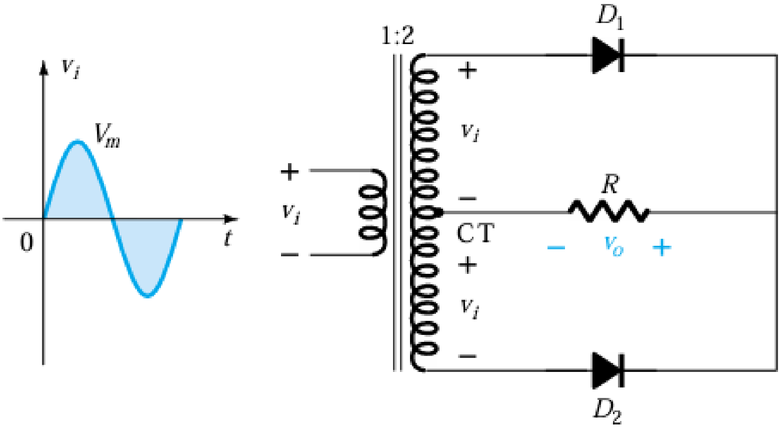
(viii) $\eta = \frac{P_{dc}}{P_{ac}}x100 = \frac{4.448}{11.524}x100 = 38.6\%$

Full Wave Rectifier

- Both the cycles of input are used.
- Two types of full-wave rectifier:
  - Center-tap rectifier
  - Bridge rectifier

Center-tap Rectifier

- Two diodes are required.
- A center-tap (CT) transformer is required to establish the input signal across each section of the secondary of the transformer.



Positive half cycle

- During the positive half cycle of the input voltage applied to the primary of the transforme, the network will appear as shown in figure.
- D_1 will be short circuit while D_2 will be open circuit. The output voltage is shown in figure.

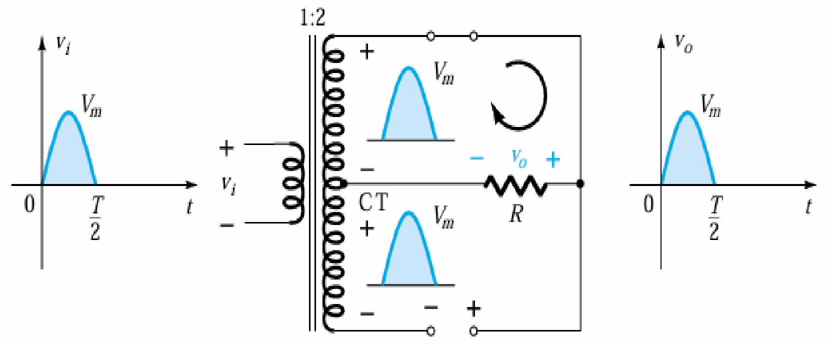


Figure 51

Negative half cycle

- During the negative half cycle of the input voltage applied to the primary of the transforme, the network will appear as shown in figure.
- D_2 will be short circuit while D_1 will be open circuit. The output voltage will maintain the same polarity across the load resistor R.

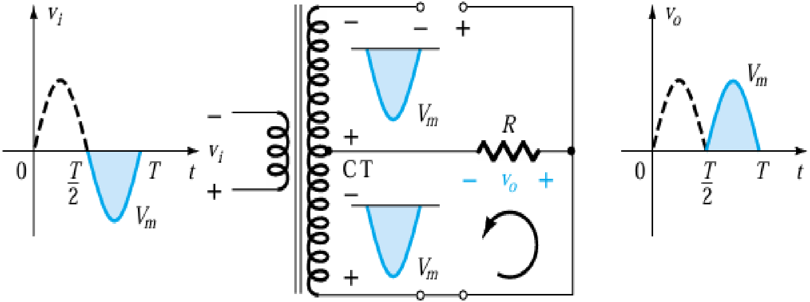


Figure 52

PIV

- The network shown below will help to determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as established by the adjoining loop will result in:
  $PIV = V_{secondary} + V_R = V_m + V_m$
  $PIV \geq 2V_m$

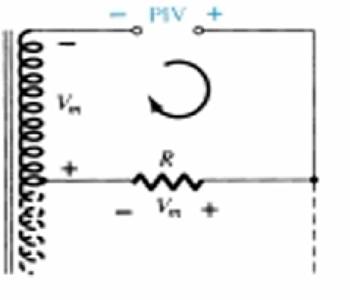


Figure 53

Bridge Rectifier

- Four diodes in a bridge configuration are used.

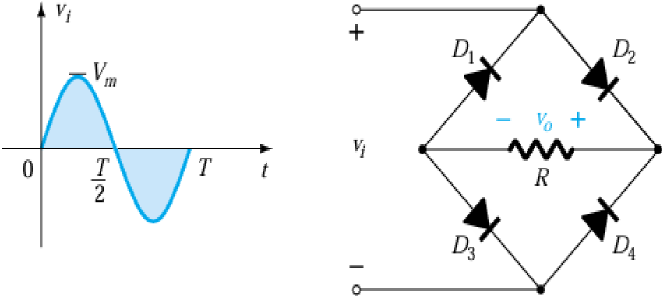


Figure 54

- During the period t= 0 to T/2 the polarity of the input is as shown in figure. The resulting polarities across the ideal diodes are also shown in figure to reveal that D2 and D3 are conducting while D1 and D4 are in the “off” state.

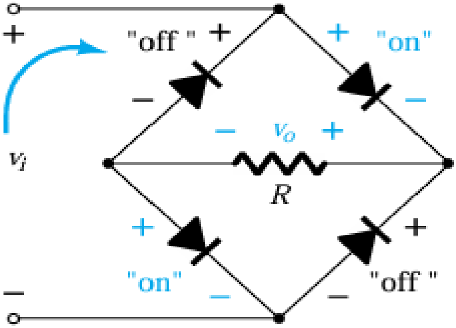


Figure 55

- Since the diodes are ideal the load voltage is $V_o = V_i$.

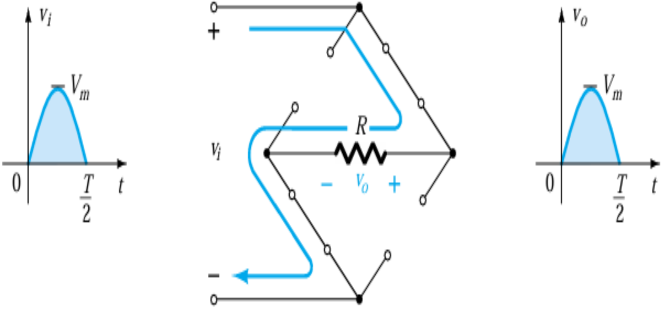


Figure 56

Negative Half Cycle

- For the negative region of the input the conducting diodes are D1 and D4, resulting in the configuration shown in figure.
- The polarity across the load resistor R is the same as for the positive half cycle, establishing a second positive pulse.

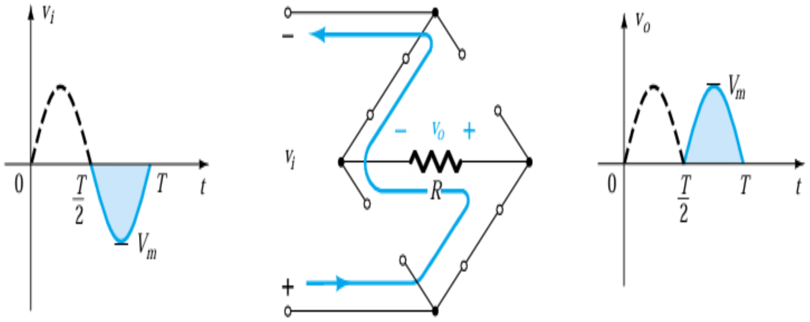


Figure 57

- Over one full cycle the input and output voltages will appear as shown in figure.

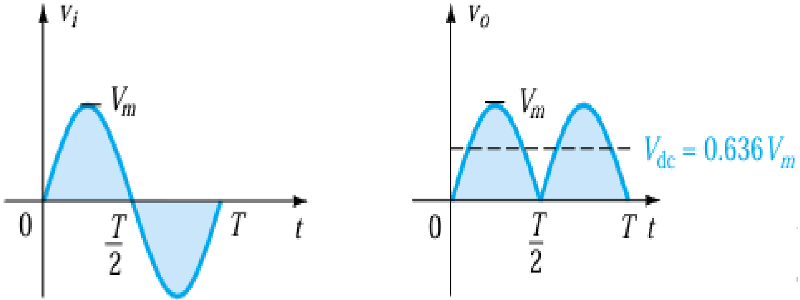


Figure 58

PIV

- The network shown below will help to determine the net PIV for each diode for this full-wave rectifier.

$PIV \geq V_m$

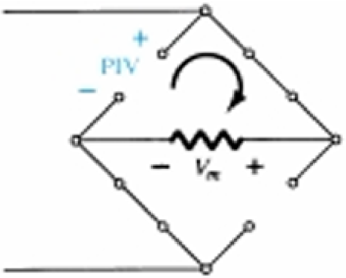


Figure 59

Circuit Analysis

- The analysis of both of the full wave rectifier circuits is the same except that • in a bridge rectifier circuit two diodes conduct during each half cycle and forward resistane becomes double i.e. $2R_F$.
  • in a bridge rectifier circuit V_{max} is the maximum voltage across the transformer secondary winding whereas in a center tap rectifier circuit V_{max} represents the maximum voltage across each half of the secondary winding.

Peak current

- The input voltage applied to the rectifier is

$$V = V_{max} sin(\omega t) \tag{43}$$

$i_1 = I_{max} sin(\omega t)$ and $i_2 = 0$ for first half cycle.
and $i_1 = 0$ and $i_2 = I_{max} sin(\omega t)$ for second half cycle.
Total current through R_L, $i = i_1 + i_2 = I_{max} sin(\omega t)$ for the whole cycle.
where,
$I_{max} = \frac{V_{max}}{R_F + R_L}$, centre tap rectifier

$I_{max} = \frac{V_{max}}{2R_F + R_L}$, bridge rectifier

DC output current

• Average or DC value of the output current can be obtained by integrating the current i_1, between 0 and π or current i_2 between π and 2π.

$I_{dc} = \frac{1}{\pi} \int_0^{\pi} i_1 \, d(\omega t)$

$I_{dc} = \frac{1}{\pi} \int_0^{\pi} I_{max} sin\omega t \, d(\omega t)$

$I_{dc} = \frac{2I_{max}}{\pi}$

DC output voltage

- Average or DC value of the output voltage across the load is given by:

$$V_{dc} = I_{dc} R_L = \frac{2}{\pi} I_{max} R_L$$

RMS value of current

$$I_{rms}^2 = \frac{1}{\pi} \int_0^{\pi} i_1^2 \, d(\omega t)$$

$$I_{rms} = \frac{1}{\pi} \int_0^{\pi} I_{max}^2 sin^2 \omega t \, d(\omega t)$$

$$= \frac{I_{max}^2}{2}$$

$$I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

RMS value of output voltage

$$V_{L_{rms}} = I_{rms} R_L = \frac{I_{max}}{\sqrt{2}} R_L$$

Rectification Efficiency

- It is defined as the ratio of dc output power to the ac input power, $\eta = \frac{P_{dc}}{P_{ac}}$

$$P_{dc} = I_{dc}^2 R_L = (\frac{2I_{max}}{\pi})^2 R_L = \frac{4}{\pi^2} I_{max}^2 R_L$$

$$P_{ac} = I_{rms}^2 (R_F + R_L) = \frac{I_{rms}^2}{2} (R_F + R_L)$$

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{8}{\pi^2} \frac{1}{1+\frac{R_F}{R_L}} = \frac{0.812}{1+\frac{R_F}{R_L}}$$

- For Bridge rectifier, $\eta = \frac{0.812}{1+\frac{2R_F}{R_L}}$

Ripple Factor

- Form factor,
  $$K_f = \frac{I_{rms}}{I_{av}} = \frac{\frac{I_{max}}{\sqrt{2}}}{\frac{2I_{max}}{\pi}} = 1.11$$
- Ripple factor,
  $$\gamma = \sqrt{K_f^2 - 1} = \sqrt{(1.11)^2 - 1} = 0.482$$

Transformer Utilization Factor

$$TUF_{primary} = \frac{P_{dc}}{P_{ac_{(rating)}}} = \frac{I_{dc}^2 R_L}{V_{rms} I_{rms}} = \frac{(\frac{2I_{max}}{\pi})^2 R_L}{\frac{V_{max}}{\sqrt{2}} \frac{I_{max}}{\sqrt{2}}} = \frac{8}{\pi^2} \cdot \frac{I_{max} R_L}{V_{max}}$$

$TUF = \frac{8}{\pi^2} \cdot \frac{R_L}{(R_F + R_L)} \approx \frac{8}{\pi^2} = 0.812$ if $R_L >> R_F$

Each secondary has TUF as 0.287 as Half wave rectifier.

$TUF_{av} = \frac{0.812 + 0.287 + 0.287}{3} = 0.693$

- For Bridge rectifier, the current flow through both primary and secondary winding are sinusoidal, hence

$TUF_{pri\&sec} = \frac{P_{dc}}{P_{ac_{(rating)}}} = \frac{I_{dc}^2 R_L}{\frac{V_{max}}{\sqrt{2}} \frac{I_{max}}{\sqrt{2}}} \approx \frac{8}{\pi^2} = 0.812$

Advantages and Disadvantages

- **Advantage:**
  - Rectification efficiency is double as compared to half wave rectifier.
  - The ripple voltage is low and of high frequency so simple filtering circuit is required.
  - Higher output voltage, higher output power and higher TUF.
- **Disadvantages:**
  - Needs more circuit elements and is costlier.

Advantages and Disadvantages of Bridge rectifiers over Center tap rectifiers

- **Advantages:**

  • No centre tap is required in the transformer secondary in bridge rectifier so tranformer required is simpler.

  • The PIV is one half that of centre tap rectifier. Thus bridge rectifier is highly suited for high voltage applications.

  • TUF is higher in bridge rectifier.

- **Disadvantages:**

  • Bridge rectifer needs four diodes, so total voltage drop in diodes become double, losses are increased and rectification efficiency is reduced.

1. A single phase full-wave rectifier uses two diodes, the internal resistance of each being 20 Ω. The transformer runs secondary voltage from centre tap to each end of secondary is 50 V and load resistance is 980 Ω. Find (i) the mean load current (ii) rms load current and output efficiency.
Ans: Given V_{rms} = 50 V, V_{max} = 50 $\sqrt{2}$ V, R_F =20 Ω, R_L= 980 Ω

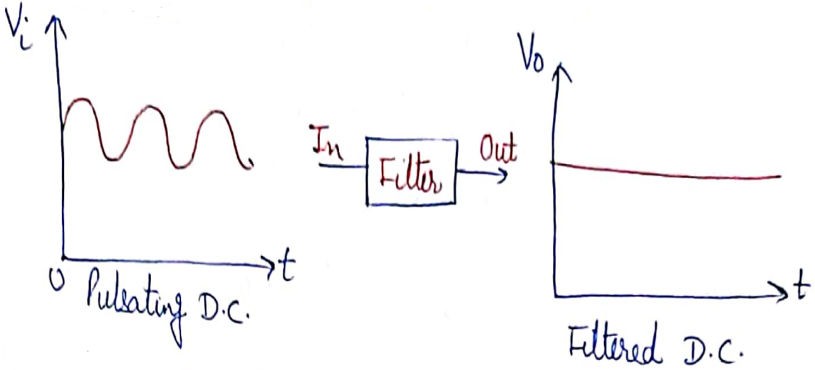
$$I_{dc} = \frac{2I_{max}}{\pi} = \frac{2V_{max}}{\pi(R_L+R_F)} = 45mA$$
$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = \frac{V_{max}}{\sqrt{2}(R_L+R_F)} = 50mA$$
$$\eta = \frac{0.812}{1+\frac{R_F}{R_L}}x100 = 79.58\%$$

2. In a center tap full wave rectifier, the load resistance R_L= 1 KΩ. Each diode has a forward bias dynamic resistance R_F= 10 Ω. The voltage across half the secondary winding is 220 sin 314t. Find (i) the peak value of current (ii)the dc output current (iii) the rms value of current (iv) the ripple factor (v) the rectification efficiency.
Ans: Given V_{max} = 220 V

$$I_{max} = \frac{V_{max}}{R_F + R_L} = 217.8mA$$
$$I_{dc} = \frac{2I_{max}}{\pi} = \frac{2x217.8}{\pi} = 138.66mA$$
$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 154mA$$
$$\gamma = \sqrt{(\frac{I_{rms}}{I_{dc}})^2 - 1} = 0.482$$
$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_F + +R_L)} = 80.26\%$$

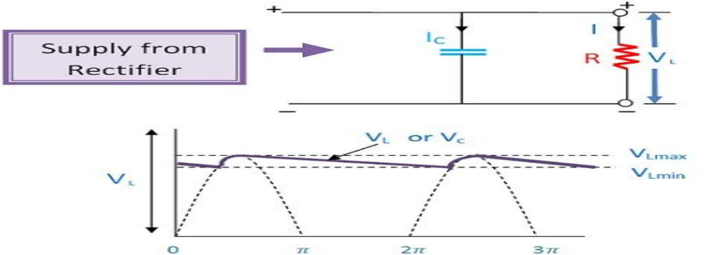
Filters

- The main function of a filter circuit is to minimize the ripple content in the rectified output.
- The output of the rectifier circuits is pulsating. It has dc and some ac components called ripples.
- This type of output is not useful for driving sophisicated electronic circuits or devices.



Shunt Capacitive Filter

- The Shunt capacitor filters comprise of capacitor along with the load resistor.
- In this, the capacitor is connected in parallel with respect to the output of rectifier circuit and also in parallel with the load resistor.
- During conduction, the capacitor starts charging and stores energy in the form of the electrostatic field.
- The capacitor will charge to its peak value because the charging time constant is almost zero.



Shunt Capacitor Filter and its Waveforms

Shunt Capacitive Filter

- During non-conduction, the capacitor will discharge through the load resistor. Thus, in this way, the capacitor will maintain constant output voltage and provide the regulated output.
- The shunt capacitor filters use the property of capacitor which blocks DC and provides low resistance to AC. Thus, AC ripples can bypass through the capacitor.
- If the value of capacitance of the capacitor is high, then it will offer very low impedance to AC and extremely high impedance to DC.
- Thus, the AC ripples in the DC output voltage gets bypassed through parallel capacitor circuit, and DC voltage is obtained across the load resistor.

Shunt Capcitive Filter

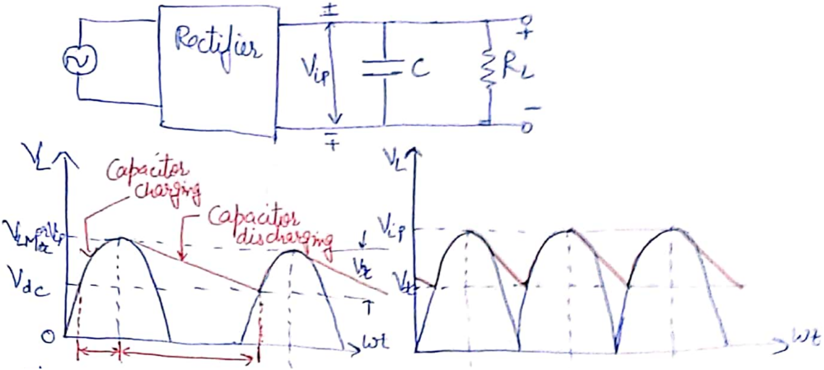


Figure 62

Shunt Capcitive Filter

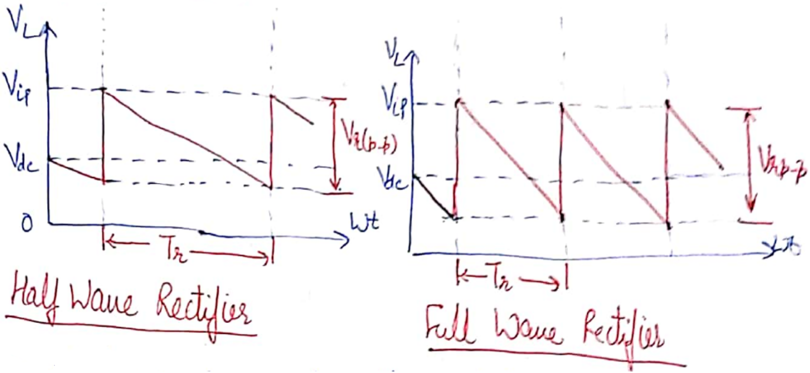


Figure 63

- The ripple voltage can be approximated by a triangular wave having peak to peak value of $V_{r(p-p)}$ and a time period of T_r centered aroundthe dc level.
- Since charging time is negligibly small, the approximate discharging time represents the full time period.
- $V_{r(p-p)}$ is the amount by which capacitor voltage falls during the discharge period T_r.
- This discharge is actually exponential ($v_C = v_{iP} e^{\frac{-t}{CR_L}}$) but can be approximated by a straight line if we assume the discharge rate to remain constant at the dc level I_{dc}.
- The charge lost dQ in time T_r is $I_{dc} T_r$.
  $V_{r(p-p)} = \frac{dQ}{C} = \frac{I_{dc} T_r}{C} = \frac{V_{dc}}{f_r C R_L}$

Calculation for Filter

- The triangular ripple has an rms value given by
$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}} = \frac{V_{dc}}{2\sqrt{3}f_r CR_L}$
$\gamma = \frac{V_{r(rms)}}{V_{dc}} = \frac{1}{2\sqrt{3}f_r CR_L}$, where γ is the ripple factor and f_r is the frequency of the ripple voltage.
- For a half wave rectifier, f_r equals the rectifier line input frequency whereas it is double the line input frequency for a full wave rectifier.
- If f is the line frequency, then
$\gamma \approx \frac{1}{2\sqrt{3}f_r CR_L}$, for half wave rectifier
$\gamma \approx \frac{1}{4\sqrt{3}f_r CR_L}$, for full wave rectifier

Calculation for Filter

- The dc voltage across the load resistor,
  $V_{dc} = V_{iP} - \frac{V_{r(p-p)}}{2}$, V_{dc} lies in mid of $V_{r(p-p)}$
  $V_{r(p-p)} = \frac{V_{dc}}{f_r CR_L}$
  $V_{dc} = V_{iP} - \frac{V_{dc}}{2f_r CR_L}$
  $V_{dc} = \frac{V_{iP}}{1+\frac{1}{2f_r CR_L}}$
  $V_{dc} = \frac{V_{iP}(2f_r CR_L)}{1+2f_r CR_L} = \frac{V_{iP}}{1+\frac{I_{dc}}{2f_r CV_{iP}}}$, for Half wave rectifier
  $V_{dc} = \frac{V_{iP}(4f_r CR_L)}{1+4f_r CR_L} = \frac{V_{iP}}{1+\frac{I_{dc}}{4f_r CV_{iP}}}$, for Full wave rectifier

1. A Half wave rectifier has a peak output voltage of 12.2 V at 50 Hz and feeds a resistive load of 100 Ω. Determine (i) the value of the shunt capacitor to give 1 % ripple factor, and (ii) the resulting dc voltage across the load resistor.

Sol: (i) $\gamma \approx \frac{1}{2\sqrt{3}f_r CR_L}$

$C = \frac{1}{2\sqrt{3}f_r R_L \gamma} = \frac{1}{2x\sqrt{3}x50x100x0.01}$

$C = 5773\mu F$

(ii) $V_{dc} = \frac{V_{iP}}{1+\frac{1}{2f_r CR_L}} = \frac{12.2}{1+\frac{1}{2x50x5773x10^{-6}x100}} = 12V$

Clippers

- There are a variety of diode networks called clippers that have the ability to "clip" off a portion of the input signal without distorting the remaining part of the alternating waveform.
- The half-wave rectifier is an example of the simplest form of diode clipper—one resistor and diode. Depending on the orientation of the diode, the positive or negative region of the input signal is "clipped" off.
- There are two general categories of clippers: series and parallel.
- The series configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in a branch parallel to the load.

Series Clippers

- Half Wave Rectifier

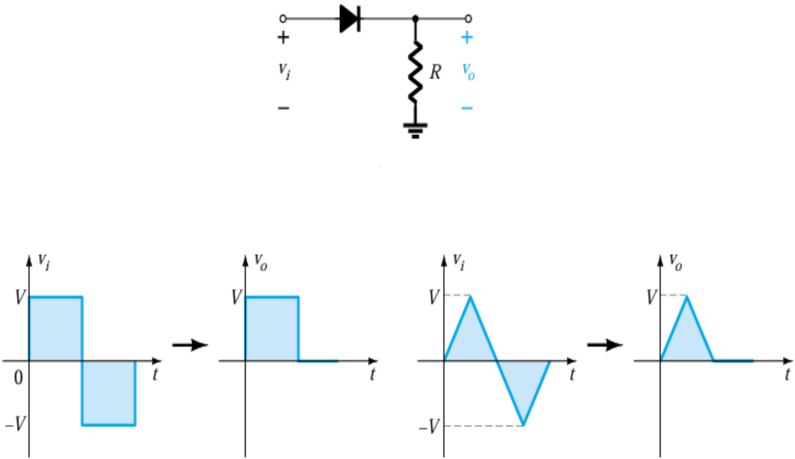


Figure 64

Series clipper with a dc supply

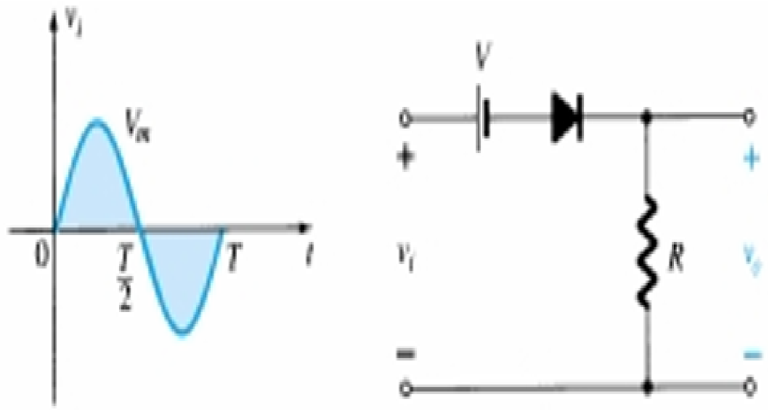


Figure 65

- Make a mental sketch of the response of the network based on the direction of the diode and the applied voltage levels.
- Determine the applied voltage (transition voltage) that will cause a change in state for the diode. For the ideal diode the transition between states will occur at the point on the characteristics where $v_d = 0$ V and $i_d = 0$ A.
  $V_i = V$

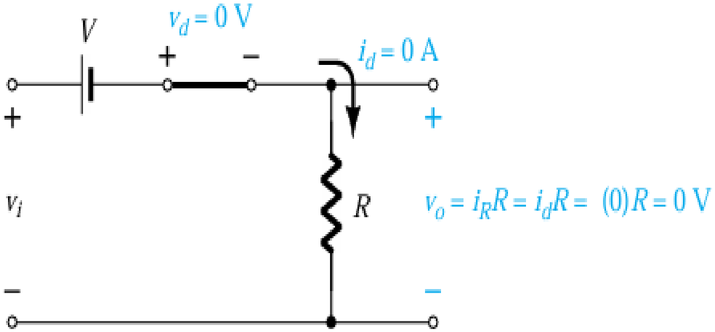


Figure 66

- Be continually aware of the defined terminals and polarity of v_o. When the diode is in the short-circuit state, such as shown in Figure, the output voltage v_o can be determined by applying Kirchhoff's voltage law in the clock wise direction:
  $v_i - V - v_o = 0$
  $v_o = v_i - V$

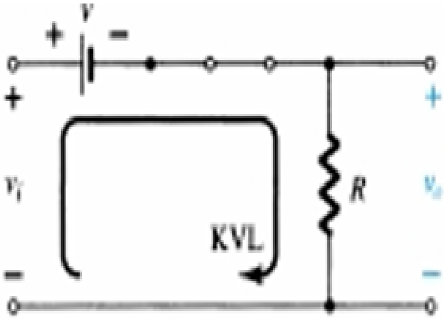
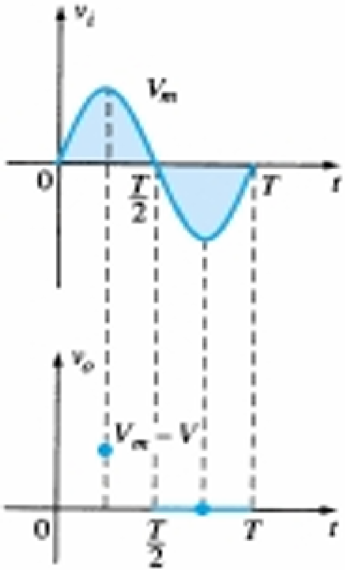


Figure 67

- It can be helpful to sketch the input signal above the output and determine the output at instantaneous values of the input.



- For instance, at $v_i = V_m$, the network to be analyzed appears in figure. For $V_m > V$ the diode is in the short-circuit state and $v_o = V_m - V$.
- At $v_i = V$ the diodes change state; at $v_i = -V_m, v_o = 0V$, the complete curve for v_o can be sketched as shown in figure.

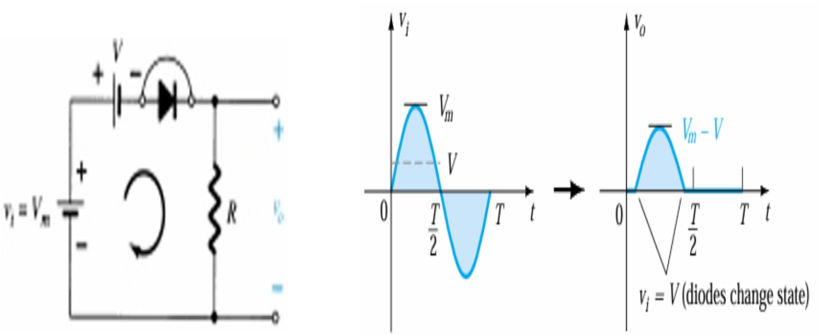


Figure 69

Series clipper with a reveresed dc supply

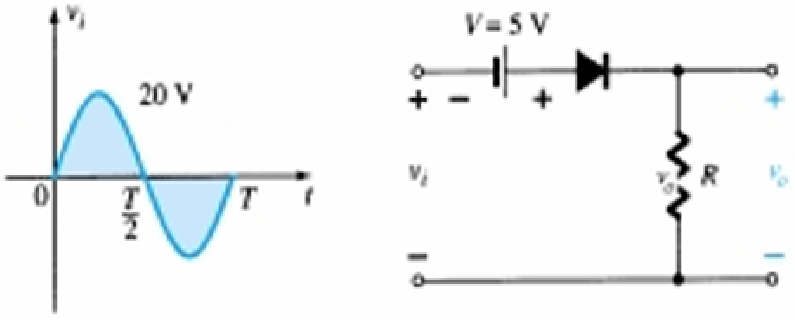
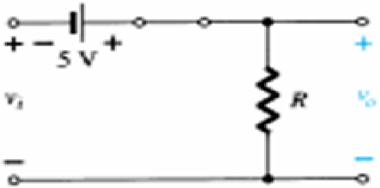
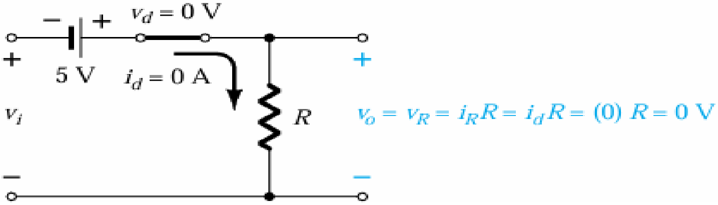


Figure 70

- The diode will be in the "on" state for the positive region of v_i, especially when we note the aiding effect of V=5 V.
  $v_o = v_i + 5$
- Substituting i_d=0 at v_d= 0 for the transistion levels, we get,
  $v_i = -5V$.





- For v_i more negative than -5 V the diode will enter its open-circuit state, while for voltages more positive than -5 V the diode is in the short-circuit state. The input and output voltages appear in figure.

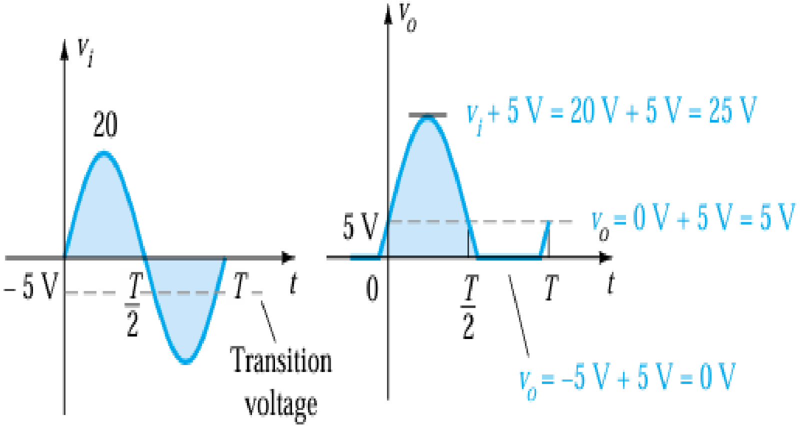


Figure 72

Square Wave Input

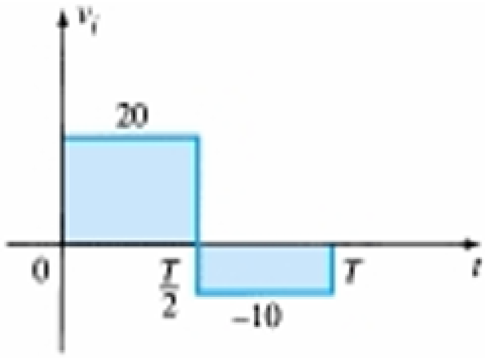


Figure 73

- When $v_i = 20$ V (0 to T/2), the diode will be in short circuit state and $v_o = 20$ V + 5 V = 25 V.
- For v_i= -10 V, the diode will be in off state and $v_o = i_R R = (0)R = 0V$.

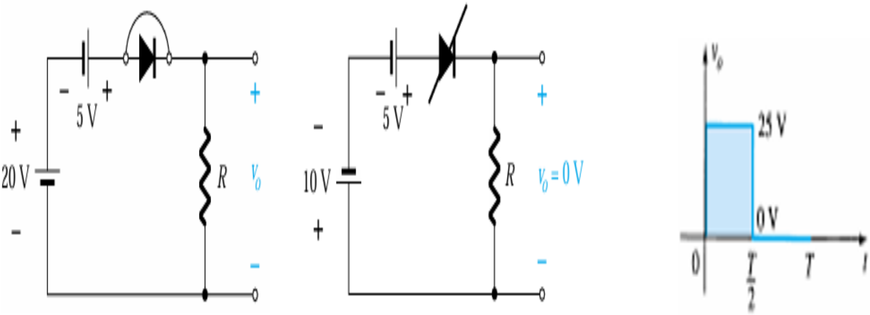


Figure 74

Parallel Clipper

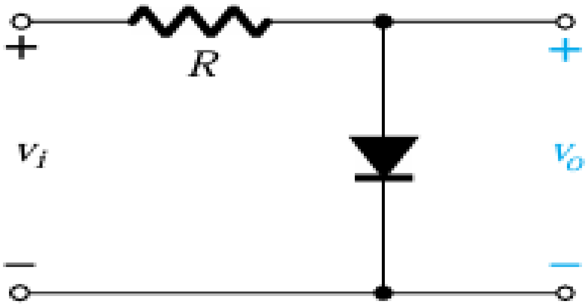


Figure 75

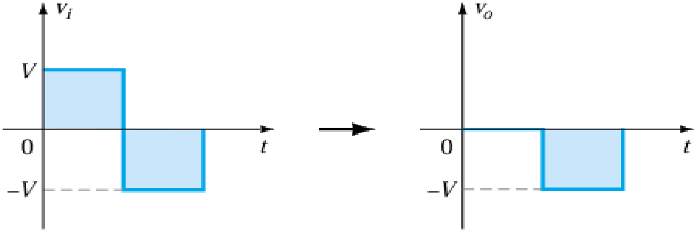


Figure 76

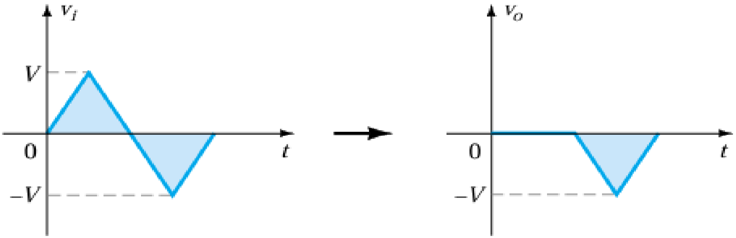


Figure 77

- Determine v_o for the given network.

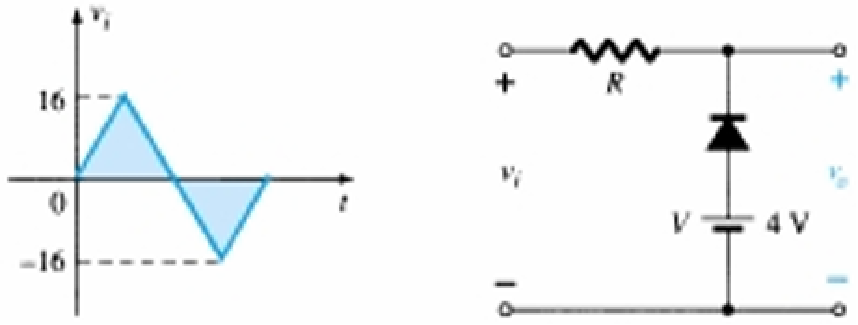


Figure 78

- The polarity of the dc supply and the direction of the diode strongly suggest that the diode will be in the "on" state for the negative region of the input signal.
- $v_o = V = 4V$

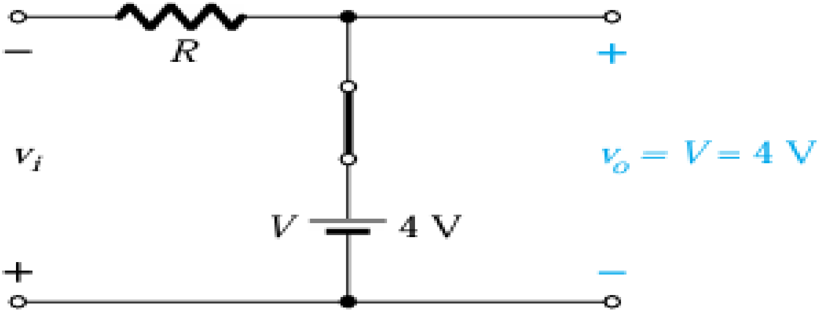


Figure 79

- The transition state can be determined from the given figure, where the condition i_d=0 A at v_d =0 V has been imposed. The result is v_i(transition)= V= 4 V.

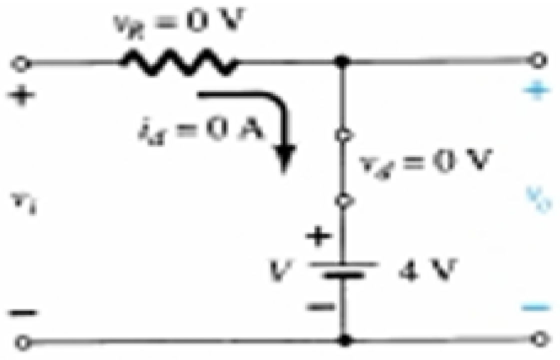


Figure 80

- Since the dc supply is obviously "pressuring" the diode to stay in the short circuit state, the input voltage must be greater than 4 V for the diode to be in the "off" state.
- Any input voltage less than 4 V will result in a short-circuited diode.
- For the open-circuit state, $v_o = v_i$.

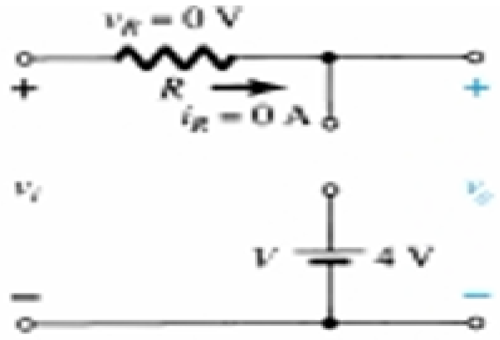


Figure 81

- The output waveform is:

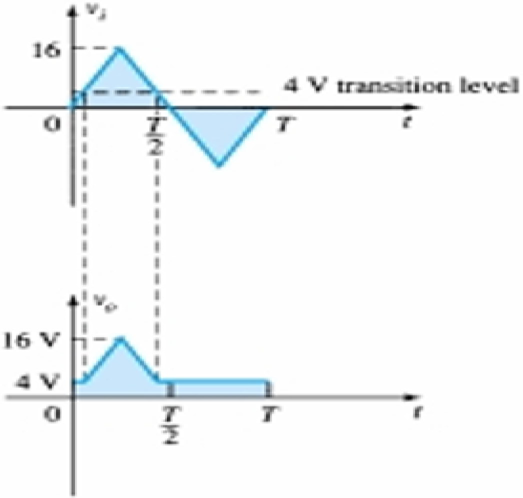


Figure 82

- Repeat previous example using a silicon diode with V_T=0.7 V.
- The transition voltage can first be determined by applying the condition i_d=0 and v_d= V_D= 0.7 V and obtaining the network given in figure.
- Applying Kirchhoff's voltage law, we find that
  $v_i + V_T - V = 0$
  $v_i = V - V_T = 4V - 0.7V = 3.3V$

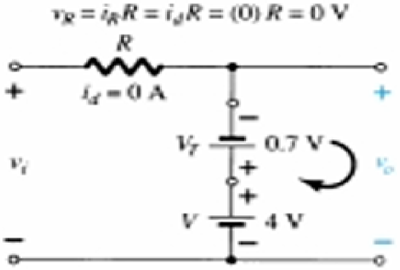


Figure 83

- For input voltages greater than 3.3 V, the diode will be an open circuit and v_i. For input voltages of less than 3.3 V, the diode will be in the "on" state,
  $v_o = 4V - 0.7V = 3.3V$

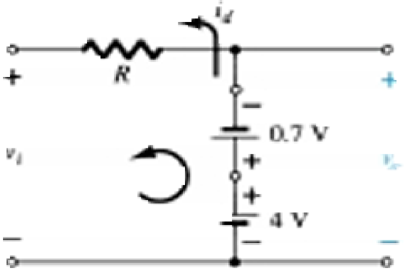


Figure 84

- The resulting output waveform is:

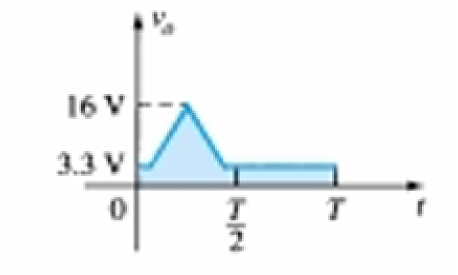


Figure 85

Simple Series Clippers (Ideal Diodes)

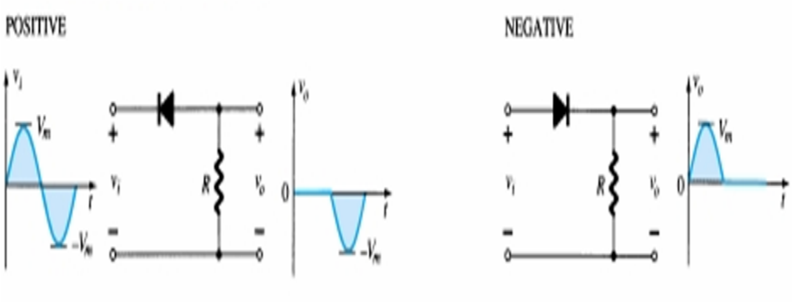


Figure 86

Biased Series Clippers (Ideal Diodes)

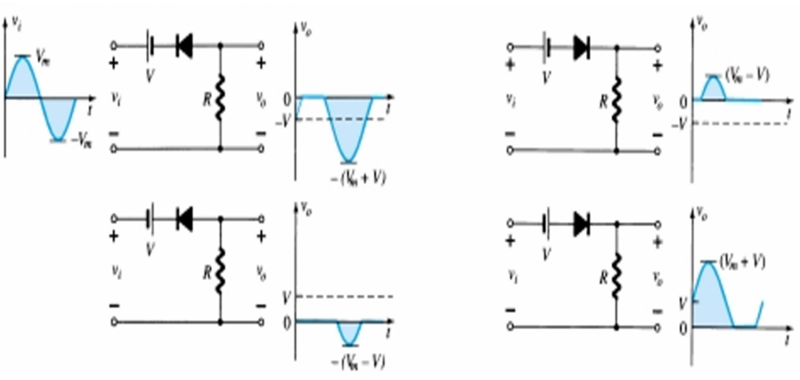


Figure 87

Simple Parallel Clippers (Ideal Diodes)

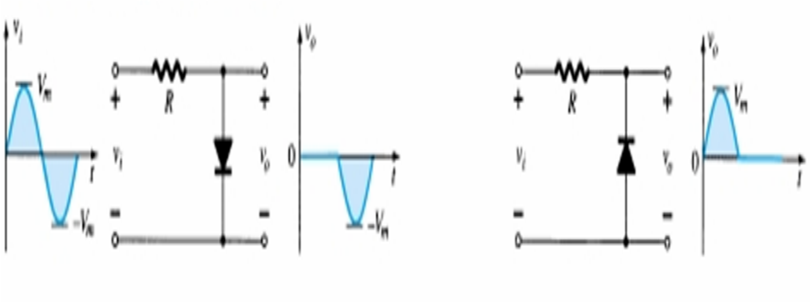


Figure 88

Biased Parallel Clippers (Ideal Diodes)

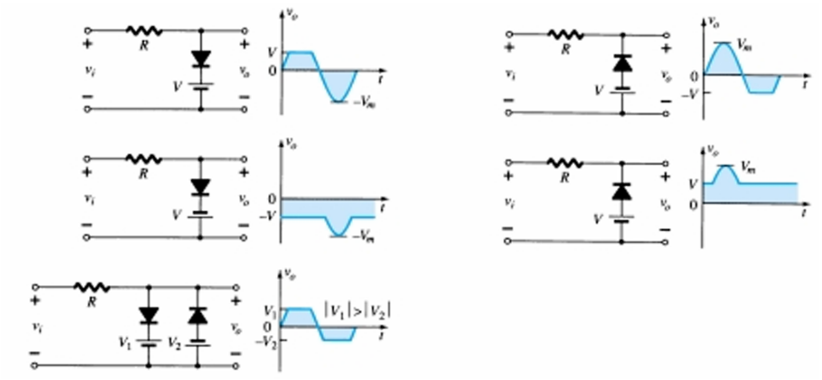


Figure 89

Clampers

- The clamping network is one that will "clamp" a signal to a different dc level.
- The network must have a capacitor, a diode, and a resistive element, but it can also employ an independent dc supply to introduce an additional shift.
- The magnitude of R and C must be chosen such that the time constant τ=RC is large enough to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is nonconducting.
- Throughout the analysis we will assume that for all practical purposes the capacitor will fully charge or discharge in five time constants.

- The network of given figure will clamp the input signal to the zero level (for ideal diodes).
- The resistor R can be the load resistor or a parallel combination of the load resistor and a resistor designed to provide the desired level of R.

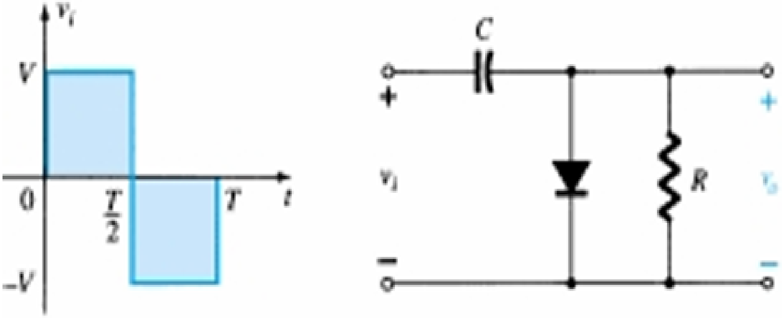
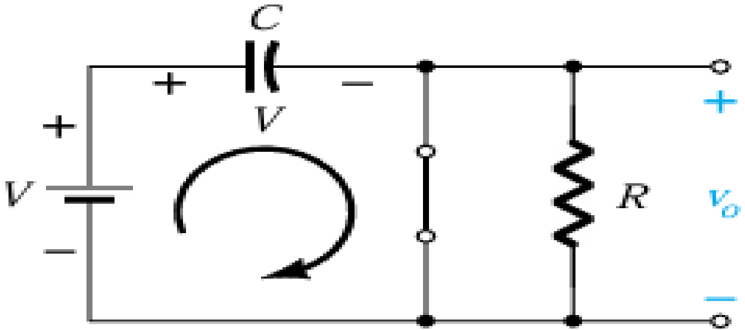
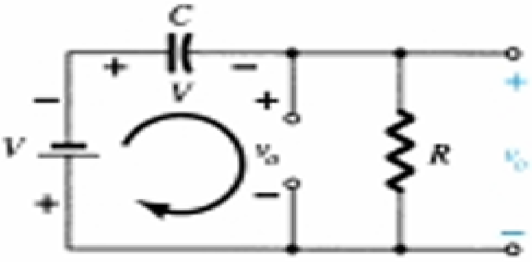


Figure 90

- During the interval 0 to T/2 the network will appear as shown in figure, with the diode in the "on" state effectively "shorting out" the effect of the resistor R.
- The resulting RC time constant is so small (R determined by the inherent resistance of the network) that the capacitor will charge to V volts very quickly.
- During this interval the output voltage is directly across the short circuit and v_o=0V.



- When the input switches to the V state, the network will appear as shown in figure, with the open-circuit equivalent for the diode determined by the applied signal and stored voltage across the capacitor—both “pressuring” current through the diode from cathode to anode.
- Now that R is back in the network the time constant determined by the RC product is sufficiently large to establish a discharge period 5τ much greater than the period $T/2 \rightarrow T$, and it can be assumed on an approximate basis that the capacitor holds onto all its charge and, therefore, voltage (since $V=Q/C$) during this period.



- Since v_o is in parallel with the diode and resistor, applying Kirchhoff's voltage law around the input loop will result in $-V - V - v_o = 0$and $v_o = -2V$
- The output signal is clamped to 0 V for the interval 0 to T/2 but maintains the same total swing (2V) as the input.

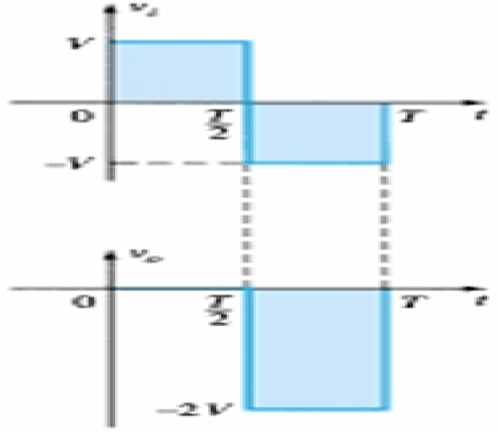


Figure 93

Analysis steps of Clamper circuits

- Start the analysis of clamping networks by considering that part of the input signal that will forward bias the diode.
- During the period that the diode is in the "on" state, assume that the capacitor will charge up instantaneously to a voltage level determined by the network.
- Assume that during the period when the diode is in the "off" state the capacitor will hold on to its established voltage level.
- Throughout the analysis maintain a continual awareness of the location and reference polarity for vo to ensure that the proper levels for vo are obtained.
- Keep in mind the general rule that the total swing of the total output must match the swing of the input signal.

- Determine v_o for the network of figure for the input indicated.

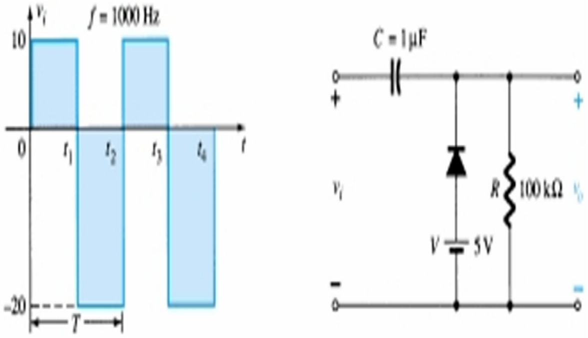
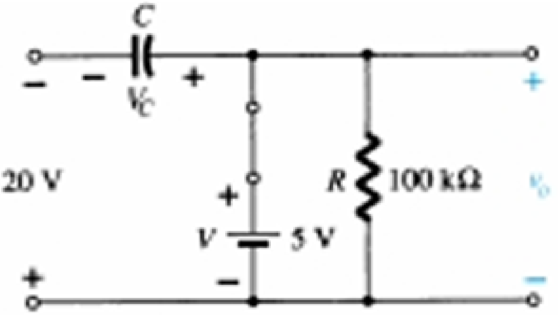


Figure 94

- The analysis will begin with the period t1 to t2 of the input signal since the diode is in its short-circuit state as recommended by comment 1.
- For this interval the network will appear as shown in figure.
- The output is across R, but it is also directly across the 5-V battery. The result is v_o =5 V for this interval.
- Applying Kirchhoff's voltage law around the input loop will result in $-20V + VC - 5V = 0$ and $V_C = 25V$. The capacitor will therefore charge up to 25 V.



- For the period t2 to t3 the network will appear as the open-circuit equivalent for the diode will remove the 5-V battery from having any effect on v_o, and applying Kirchhoff's voltage law around the outside loop of the network will result in
  $+10V + 25V - v_o = 0$
  $v_o = 35V$

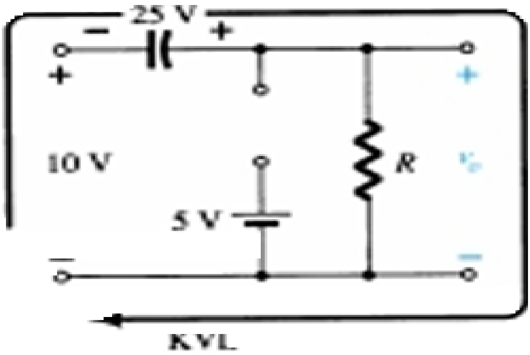


Figure 96

- Time constant - τ = RC=100 kΩ x 0.1 μF= 10 ms.
- Total discharging time is therefore 5τ = 50 ms.

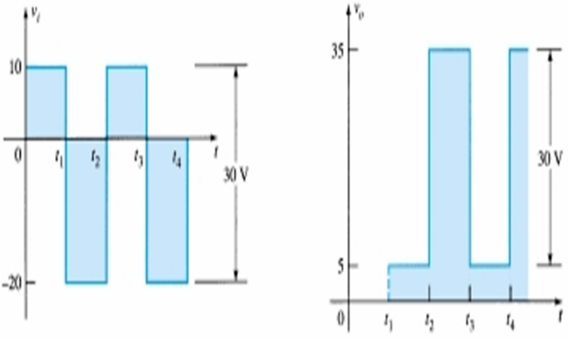
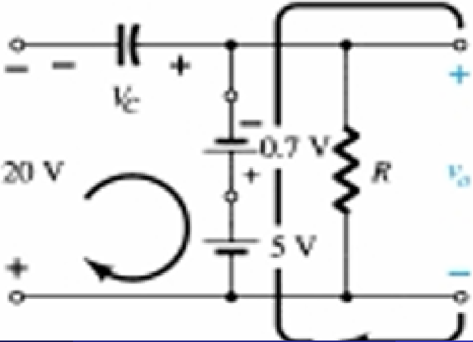


Figure 97

- Repeat previous example using a silicon diode with $V_T = 0.7$ V.
- For the short-circuit state the network now takes on the appearance of given figure and v_o can be determined by Kirchhoff's voltage law in the output section.
  $+5V - 0.7V - v_o = 0$
  $v_o = 5V - 0.7V = 4.3V$
- For the input section Kirchhoff's voltage law will result in
  $-20V + V_C + 0.7V - 5V = 0$
  $V_C = 25V - 0.7V = 24.3V$



- For the period t2 to t3 the network will now appear as in given figure, with the only change being the voltage across the capacitor. Applying Kirchhoff's voltage law yields:
  $+10V + 24.3V - v_o = 0$
  $v_o = 34.3V$

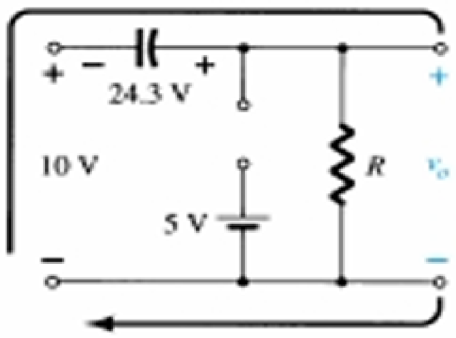


Figure 99

- The resulting output appears in figure, verifying the statement that the input and output swings are the same.

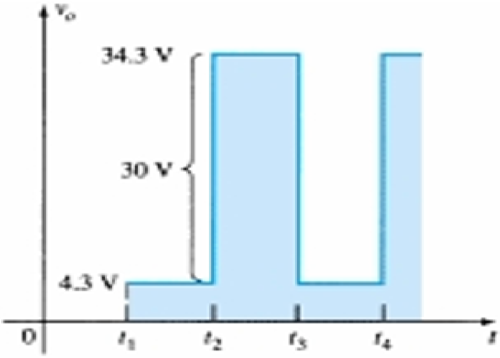


Figure 100

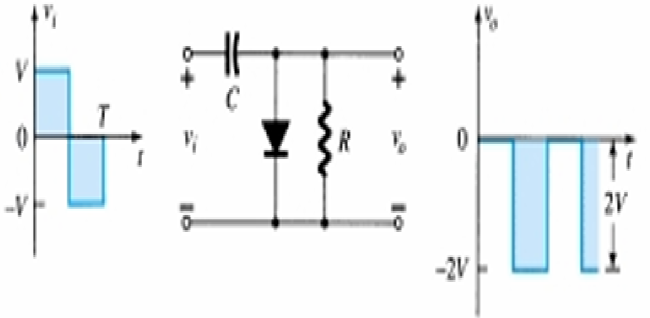


Figure 101

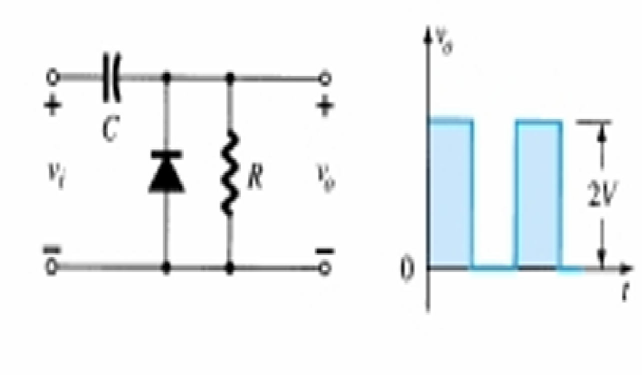


Figure 102

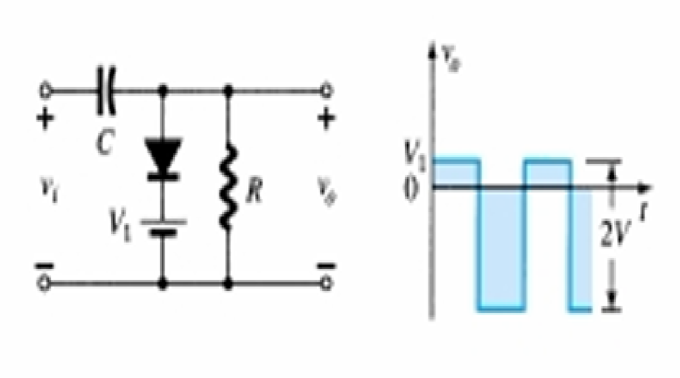


Figure 103

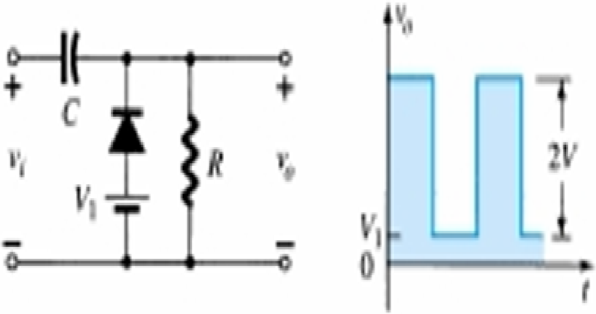


Figure 104

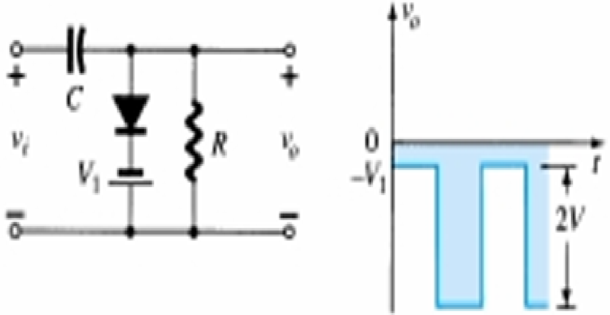


Figure 105

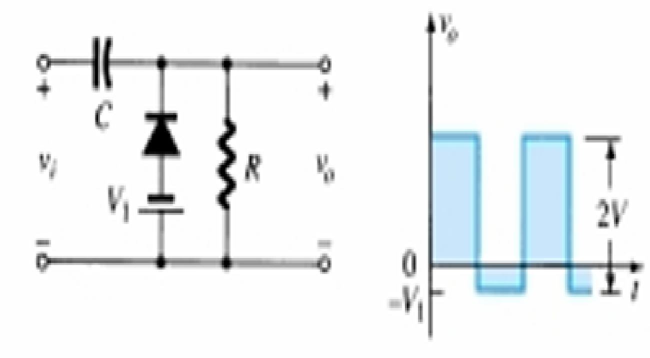


Figure 106

Clamping network with a sinusoidal input

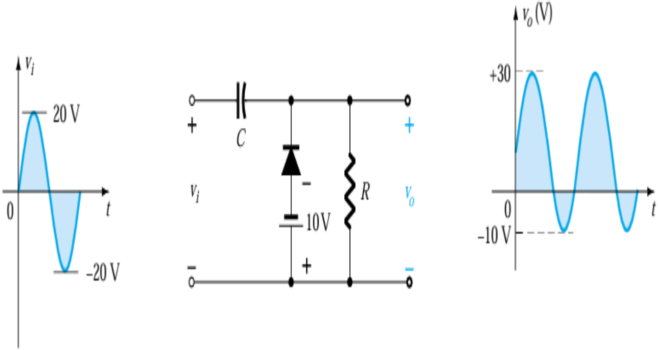


Figure 107

Special-Purpose Diodes

- Light-emitting diodes

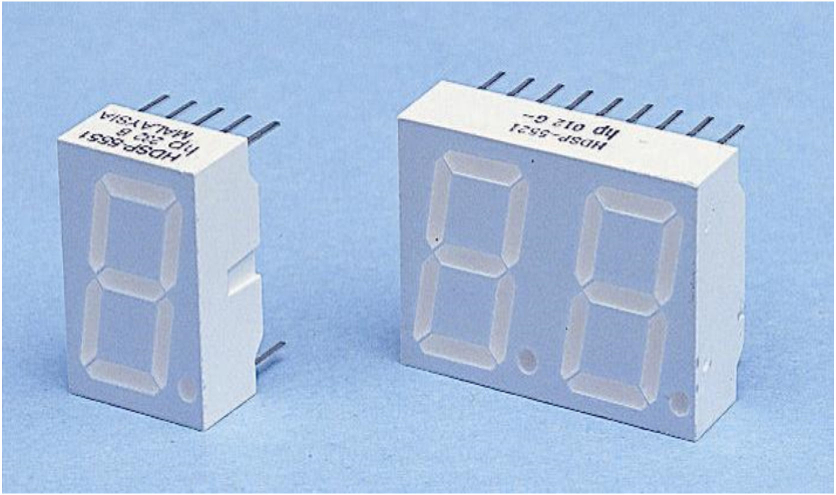


Figure 108

Special-Purpose Diodes

• Zener Diodes -uses the relatively constant reverese breakdown voltage to produce a voltage reference breakdown voltage is calle the Zener voltage, V_Z - output voltage of circuit shown is equal to V_Z despite variations in input voltage V - a resistor is used to limit the current in the diode

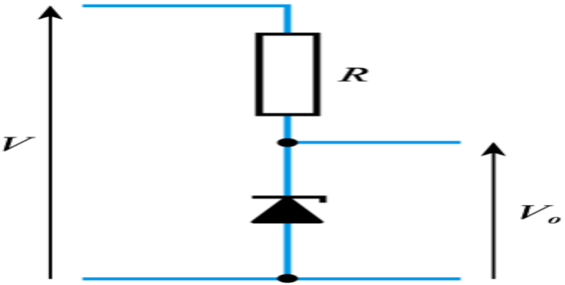


Figure 109

Special-Purpose Diodes

- Schottky diodes

- formed by the junction between a layer of metal (e.g. aluminium) and a semiconductor
- action relies only on majority charge carriers
- much faster in operation than a pn junction diode
- has a low forward voltage drop of about 0.25 V
- used in the design of high-speed logic gates

Special-Purpose Diodes

- Tunnel diodes

- high doping levels produce a very thin depletion layer which permits 'tunelling' of charge carriers
- results in a characteristic with a negative resistance region
- used in high-frequency oscillators, where they can be used to 'cancel out' resistance in passive components

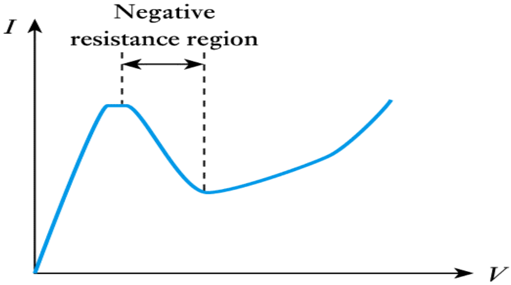


Figure 110

Special-Purpose Diodes

- Varactor diodes

- a reversed-biased diode has two conducting regions separated by an insulating depletion region
- this structure resembles a capacitor
- variations in the reverse-bias voltage change the width of the depletion layer and hence the capacitance
- this produces a voltage-dependent capacitor
- these are used in applications such as automatic tuning circuits

END OF MODULE 1